

Theory of Computing:

10. Turing Machine - 1



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Outline :

- TM Architecture
- Examples
 - $\{w \mid w \text{ in the form } 0^*1^*\}$
 - $\{w \mid w \text{ contains } 101\}$
 - $\{0^n1^n \mid n \geq 0\}$
 - $\{a^n b^n c^n \mid n \geq 0\}$
 - $\{w\#w \mid w \text{ in } \{0,1\}^*\}$
 - $\{1^n \times 1^m = 1^{n+m}\}$
- Formalism for TM
- Classes of Languages

Computer Science & Programming

- Programming variables, computer science....

Chomsky Classification of Languages

- Lastly, Turing Machine will be explained next week

Type	Grammar	Language	Automaton
Type-3	Regular Grammar	Regular Languages	DFA/NFA
Type-2	Context-Free Grammar	Context Free Languages	PDA
Type-1	Context-Sensitive Grammar	Context-Sensitive Languages	Linear-bounded automaton
Type-0	Unrestricted grammar	Recursively enumerable language	Turing Machine

TM Architecture : Introduction

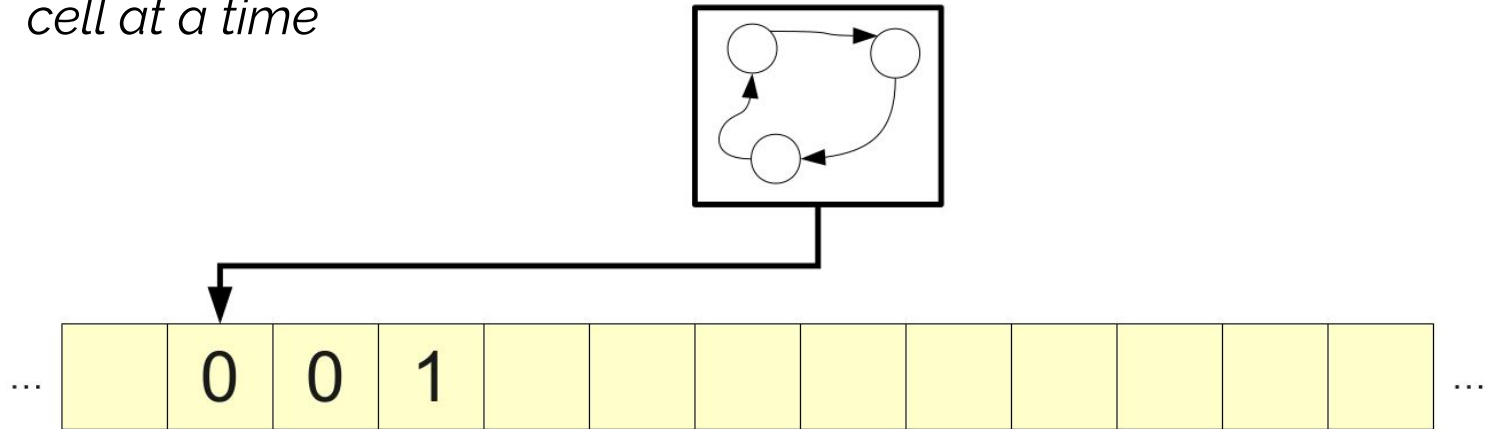
- Alan Turing aimed to design a computational machine which is :
 - Simple
 - Intuitive
 - Generic and
 - Formalizes the computation performed by a **human mind**

TM Architecture : Introduction

- Turing Machine was introduced in by its a British mathematician Alan Turing in 1936
- Turing machine is a much more accurate model of a general purpose computer with almost the same power.
 - It can execute any algorithm.

TM Architecture : Definition

- *A Turing machine is a finite automaton equipped with an infinite tape as its working memory.*
- *The machine has a tape head that can read and write a single memory cell at a time*



TM Architecture : Components

- A Turing Machine (TM) has three components:
 - An infinite tape divided into cells.
 - Each cell contains one symbol.
 - By Default, all cells are filled with the Blank Symbol : $\sqcup$
 - The input string is placed on the tape at the left side.
 - Other symbols not from the language alphabet can be written to the tape
 - By default, the tape is infinite from the right side.

1

TM Architecture :

Components

- A Turing Machine (TM) has three components:
 - A head that accesses one cell at a time:
 - It can both read from and write on the tape
 - It can move both left and right (based on the transitions)

2

TM Architecture :

Components

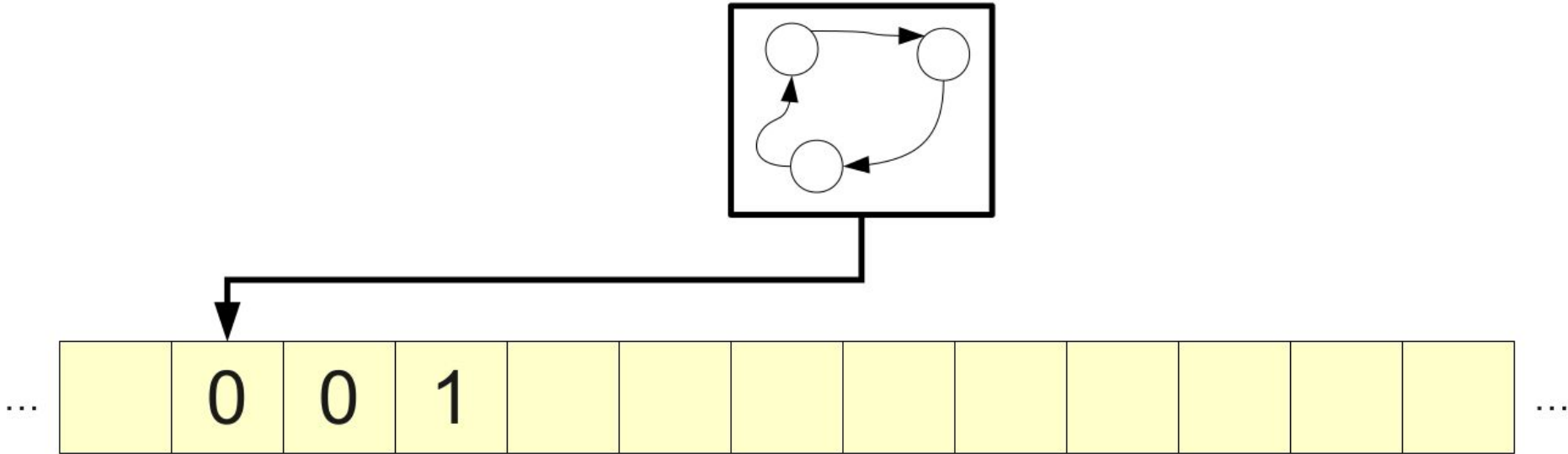
- A Turing Machine (TM) has three components:
 - A program memory or Controller for issuing commands :
 - Finite number of states
 - The transitions between the states

3

TM Architecture : Execution

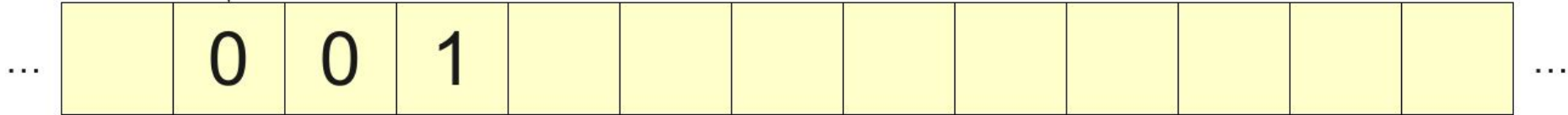
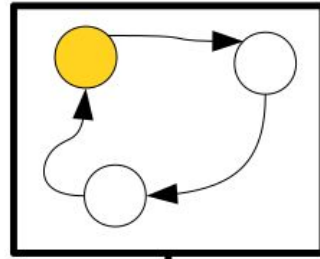
- At each step , the Turing Machine :
 1. Read the cell symbol from the Tape
 2. Decides which transition to make
 3. Write a Symbol to tape cell under the current tape head
 4. Changes the state
 5. Moves the tape head to the left or to the right.

TM Architecture : Execution & Simulation



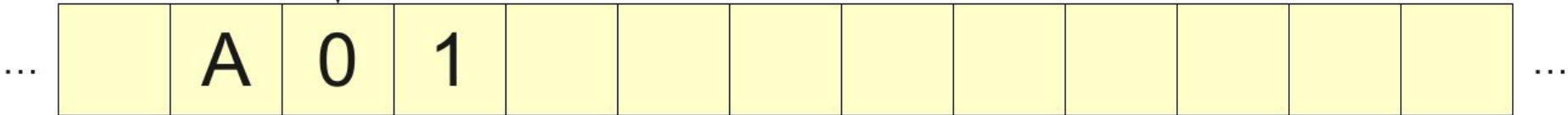
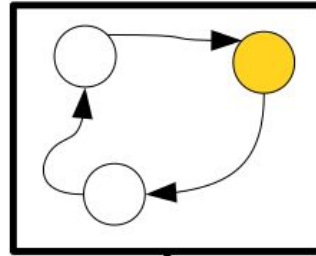
TM Architecture : Execution & Simulation

1. The start state is selected
2. The symbol is read under the current tape head
3. The relevant transition is selected

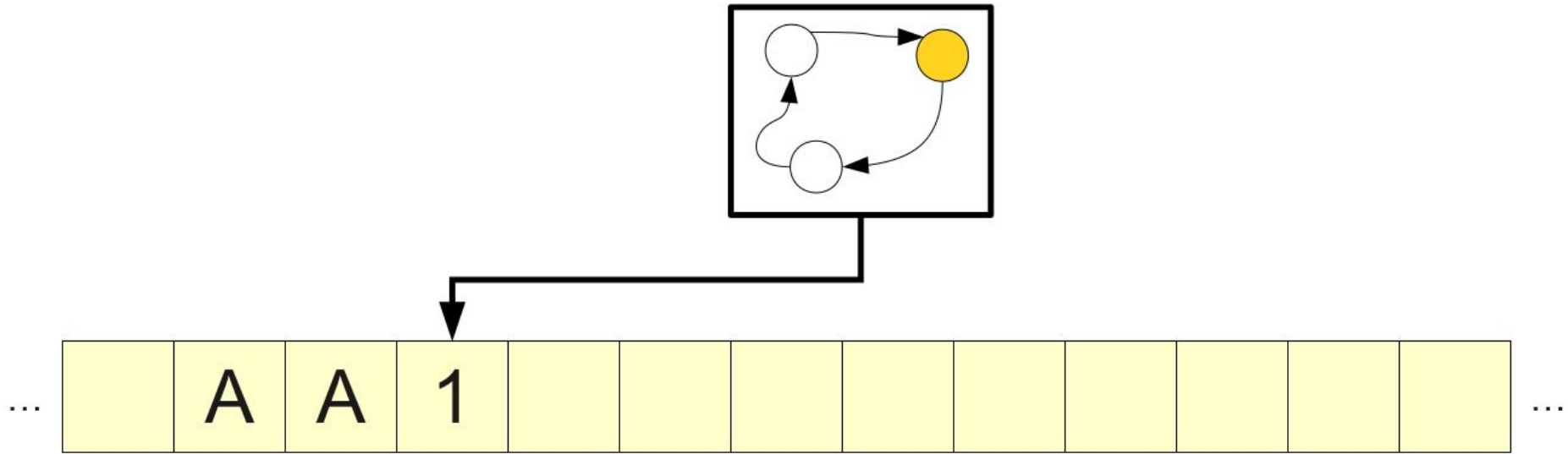


TM Architecture : Execution & Simulation

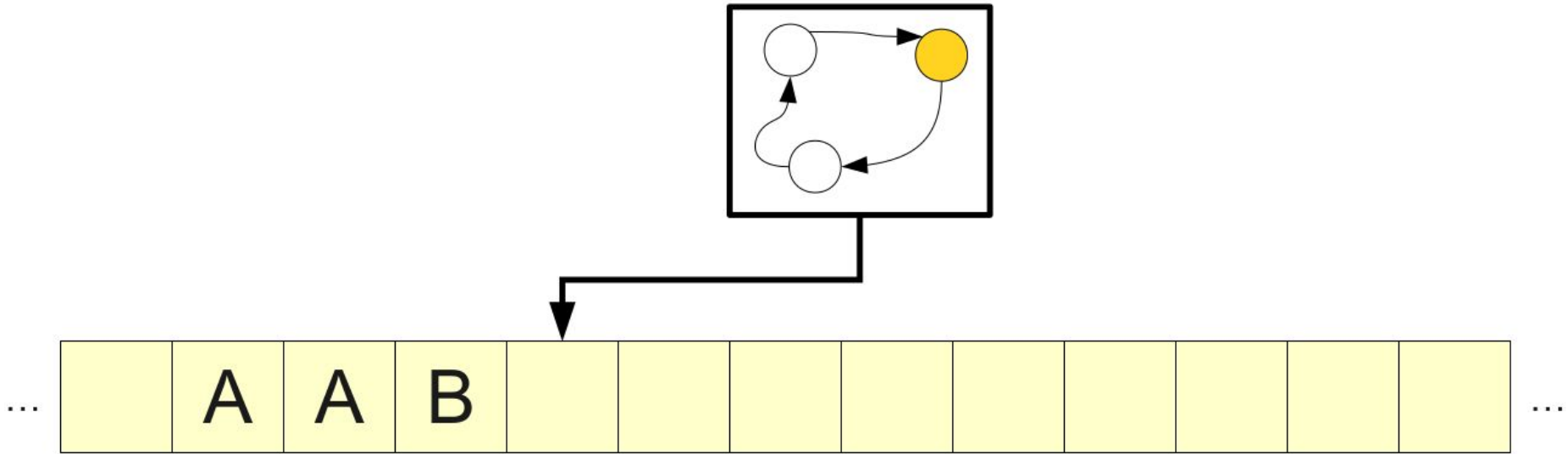
4. A is written in the first Cell.
5. State is updated
6. Tape head is moved to the right



TM Architecture : Execution & Simulation



TM Architecture : Execution & Simulation



TM Architecture : Execution & Simulation

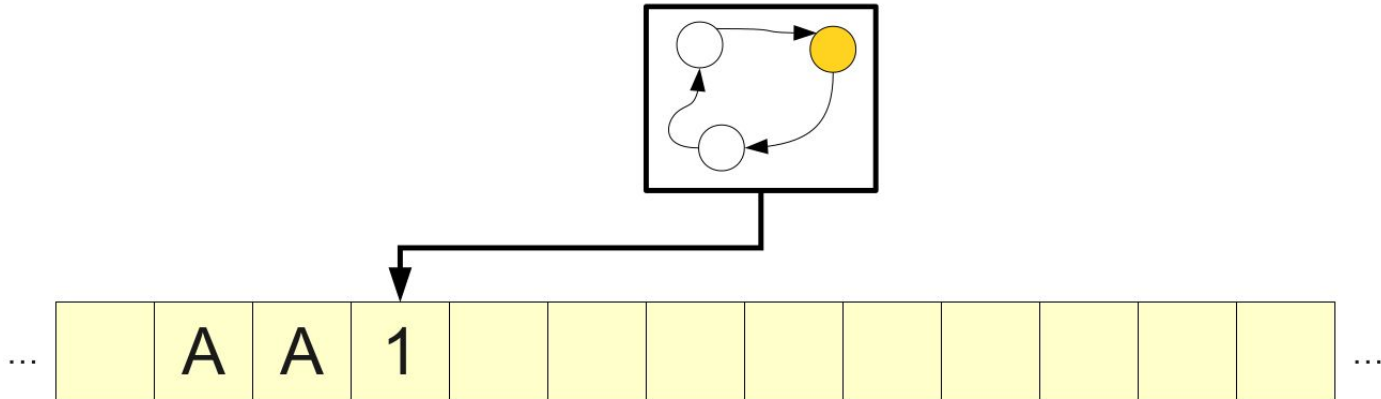


Move Right



TM Architecture : Transitions

- Turing Machine is represented as a diagram like Finite State Machine
 - Except that each arrow is labeled with the following format:
 - $A \rightarrow B, R$: When you read A , replace it with B and move **right**



TM Architecture : Transitions

- Turing Machine is represented as a diagram like Finite State Machine
 - Except that each arrow is labeled with the following format:
 - $A \rightarrow B, R$: When you read A, replace it with B and move Right
 - $A \rightarrow B, L$: When you read A, replace it with B and move Left
 - $A \rightarrow R$: When you read A, replace it with A and move Right
 - $A \rightarrow L$: When you read A, replace it with A and move Left

TM Architecture :

Transitions : Notations

- Depending on the textbook or lecture notes:
 - Transition : $a \rightarrow$: means when you read a , move right.
 - Transition : $\triangleright \rightarrow$: means when you are at the start of the tape from the left side, move right. Assumption that the first cell on the left contain the $\triangleright$ symbol
 - Transition : ABR : when you read A , replace it with B , and move Right
 - Different notations for the blank symbol : $\sqcup$ or $\square$ or Δ
 - There are other textbooks claiming the direction **S** symbol to stay in place ??

TM Architecture : States

- Turing machine has the following types of states
 - Single Start State
 - Intermediate States
 - Accept State : with the label as Accept and drawn in double lines
 - Reject State : with the label : reject , single line

TM Architecture : States

- Turing machine has the following types of states
 - Single Start State
 - Intermediate States
 - Accept State
 - Reject State : At any state if there is no relevant transition, it means, there is implicitly a transition to the reject state.

TM Architecture :

Accepting vs. Rejecting vs ...

- The output for a Turing machine for a given input strings:
 - Halts
 - Keeps looping without halting :
 - TM keeps finding valid transitions between states even for a smaller input string

TM Architecture :

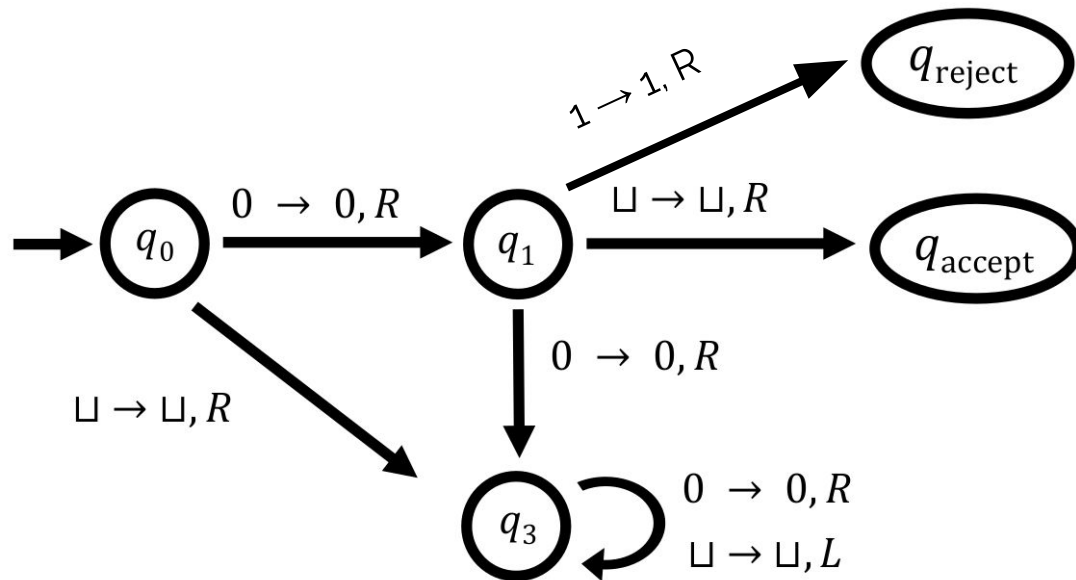
Accepting vs. Rejecting vs ...

- The output for a Turing machine for a given input strings:
 - Halts
 - Accept State : the word is accepted to be part of the language
 - Reject State : the word is not accepted within the language
 - Keeps looping without halting:
 - Undecidable

TM Architecture :

Accepting vs. Rejecting vs ...

- What does this TM do in input : **000**

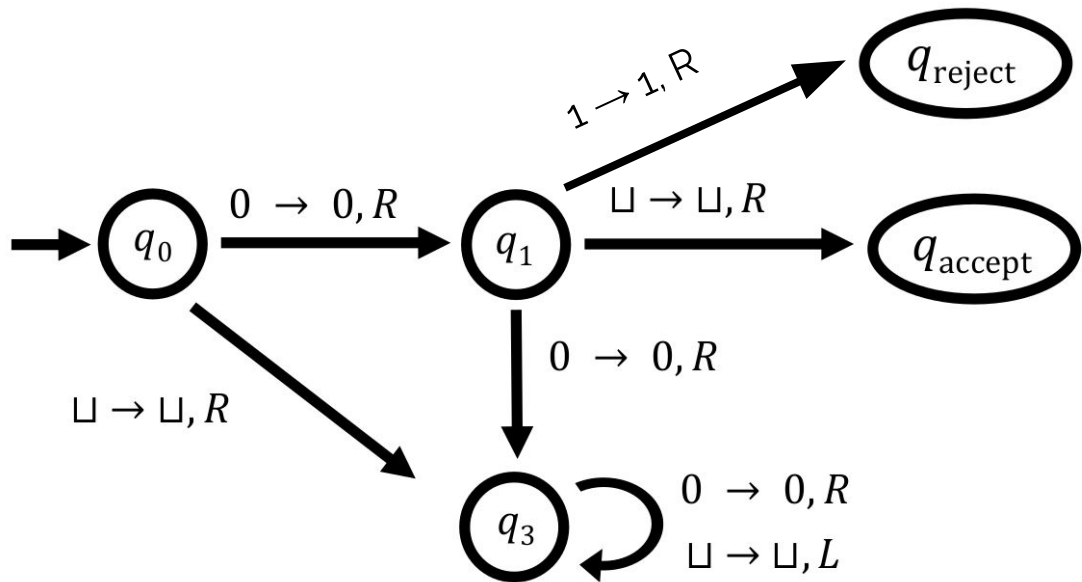


TM Architecture :

Accepting vs. Rejecting vs ...

- What does this TM do in input : **000**

- Halt and Accept
- Halt and Reject
- Halt in State q_3
- Loop Forever



TM Architecture : Machine Configuration

- At each step, the machine would have a configuration reflecting :
 - Current state
 - Position of the Tape Head
 - Content of the Tap

TM Architecture : Machine Configuration

- At each step, the machine would have a configuration reflecting :
 - Sipser prints the symbols in the tape whilst show the current state just before the head, Examples
 - $q_1 0000$: Head at State Q_1 at the beginning of the tape
 - $\sqcup y q_5 x x \sqcup$: Tape contains two cells containing $\sqcup$ and y , the tape head points at **x** whilst the current state is **q_5**

TM Architecture : Machine Configuration

- A preferable way is to show the transitions table , whilst each row represents a configuration

Time	Configuration	State	Tape						
0	C_0	q_0	▷	b	b	b	□	□	...
1	C_1	q_1	▷	b	b	b	□	□	...
2	C_2	q_4	▷	b	b	b	□	□	...
3	C_3	q_4	▷	b	b	b	□	□	...
4	C_4	q_4	▷	b	b	b	□	□	...
5	C_5	q_{rej}	▷	b	b	b	□	□	...

TM Architecture : Construction

- Creating and Innovative Process
 - Describe in English the algorithm containing the instructions :
 - How to move the head
 - What to write on the tape
 - Visualize your algorithm with the state diagram

TM Architecture : Construction

- Very important assumption :
 - Given words to be processed by a Turing machine, should never contain blank “space”

Examples for TM :

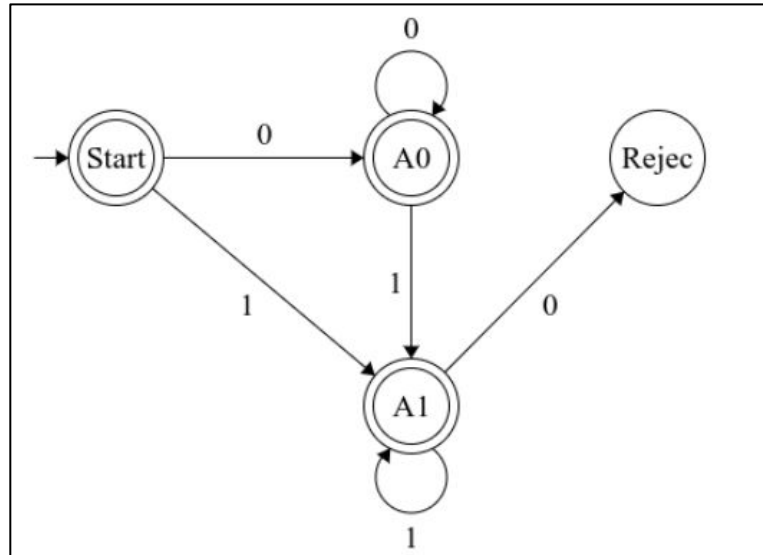
0^*1^*

- The DFA for the language 0^*1^* is given as :

Examples for TM :

0^*1^*

- The DFA for the language 0^*1^* is given as :



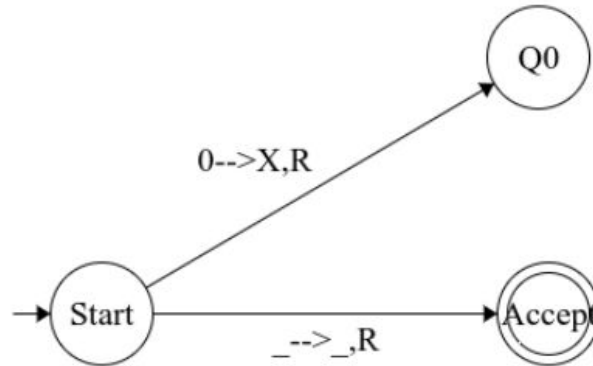
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



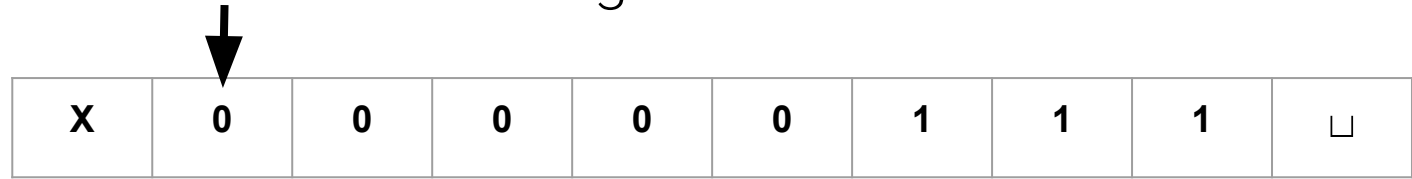
Start with the simple case
When the string is empty
Or : when first letter is 0



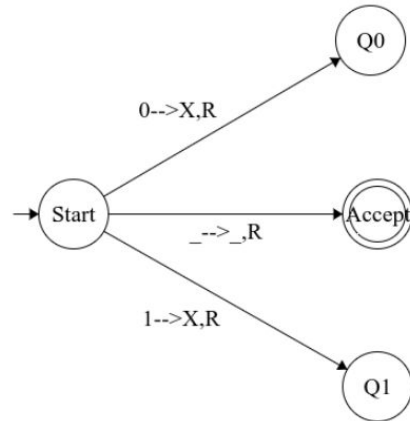
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



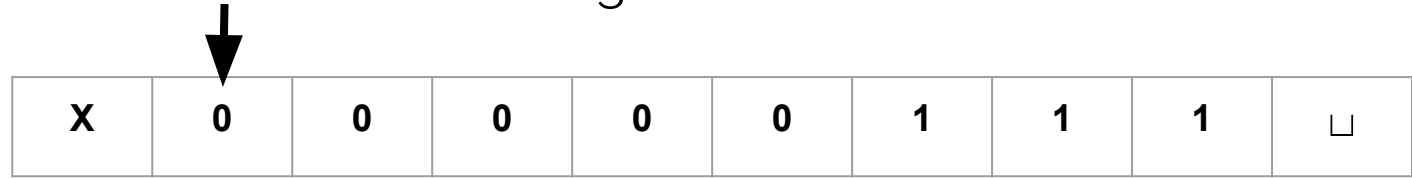
What if first letter is 1 ?



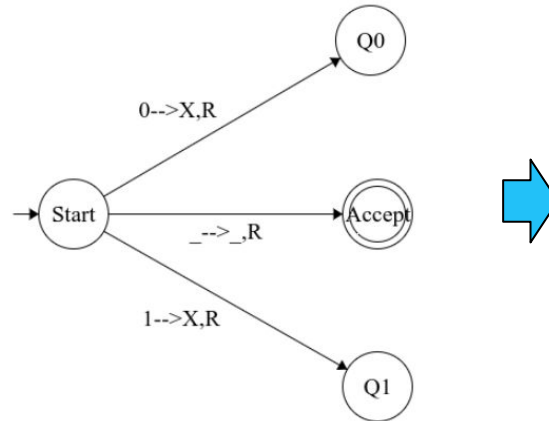
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



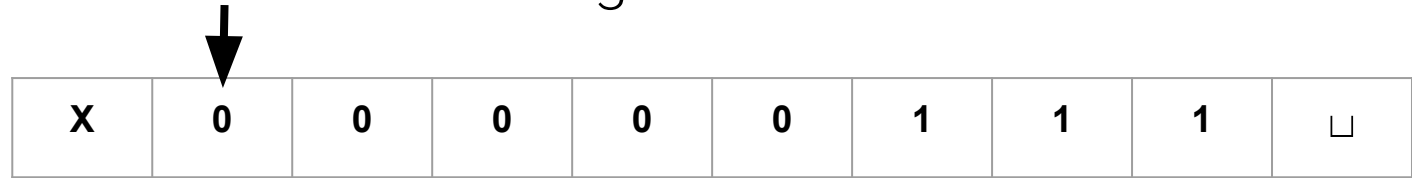
If input string is only : 0



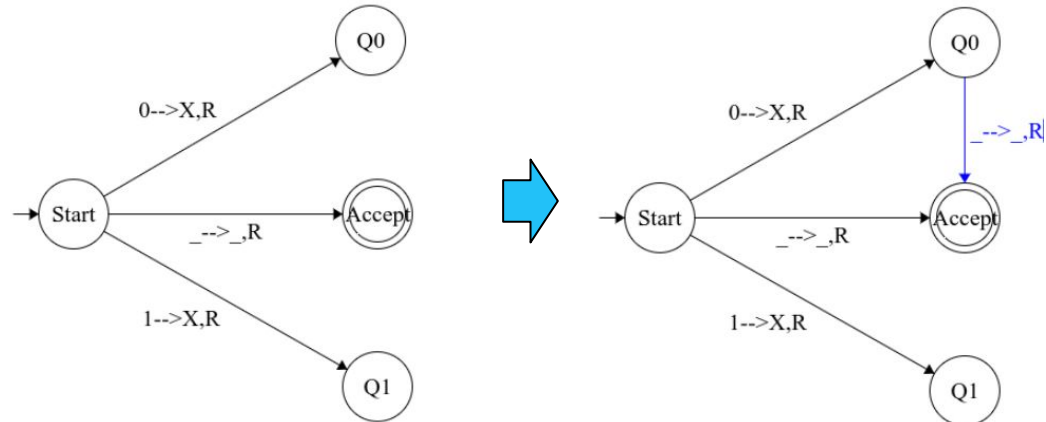
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



If input string is only : 0



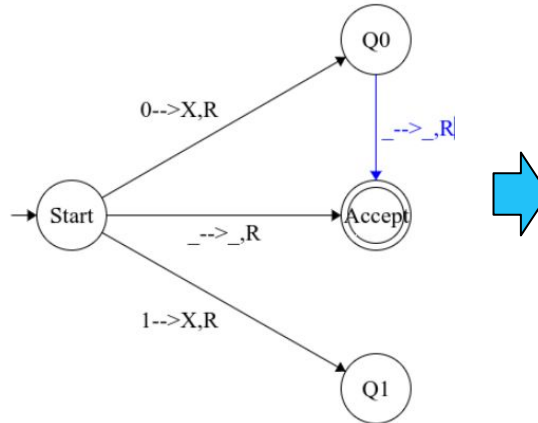
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



If input string is : 000...00



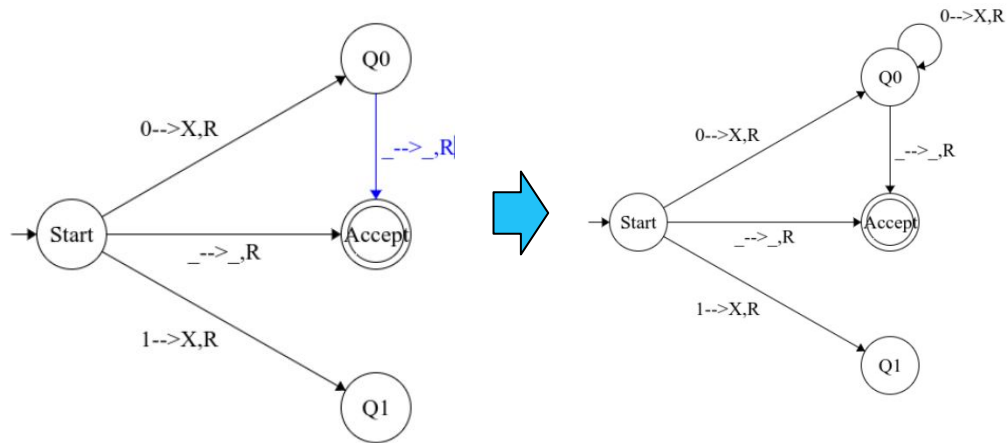
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



If input string is : 000...00



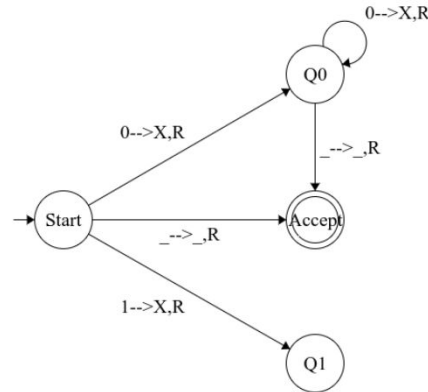
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



If input string is : 00...01



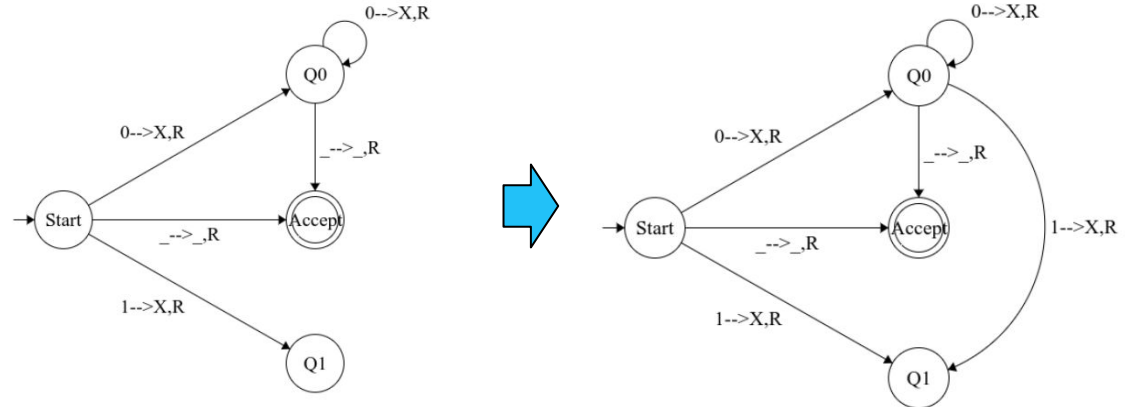
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



If input string is : 00...01



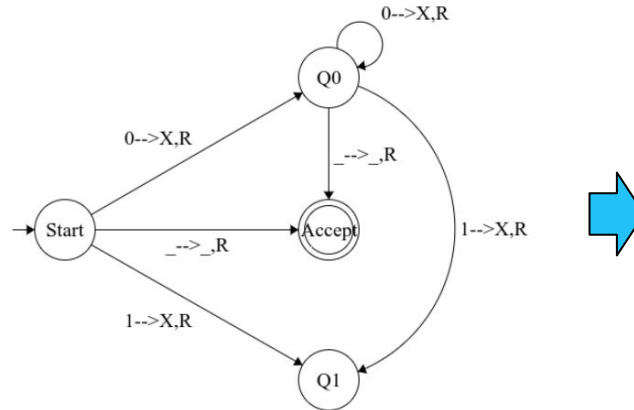
Examples for TM :

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- For the finite automaton of the Turing Machine :



TM needs to terminate at
Accept



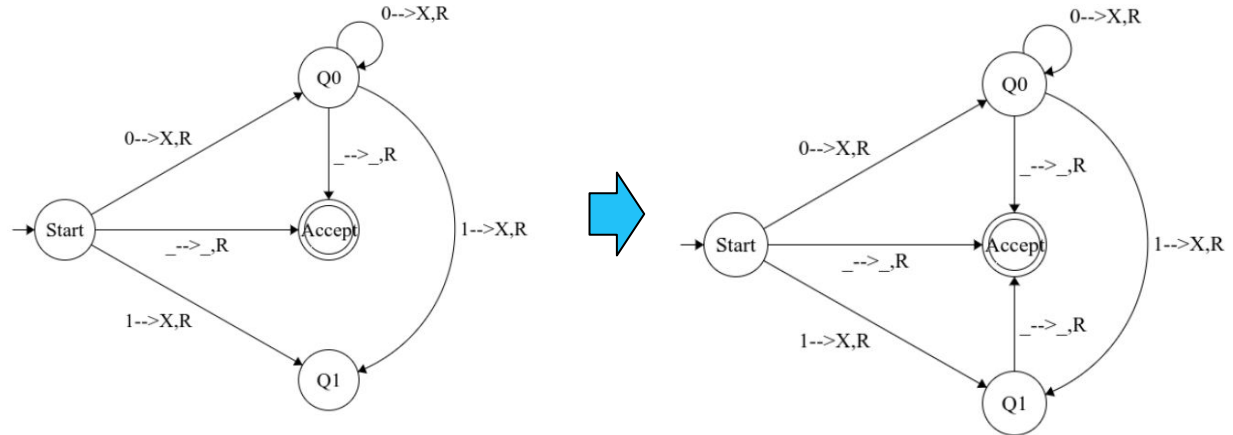
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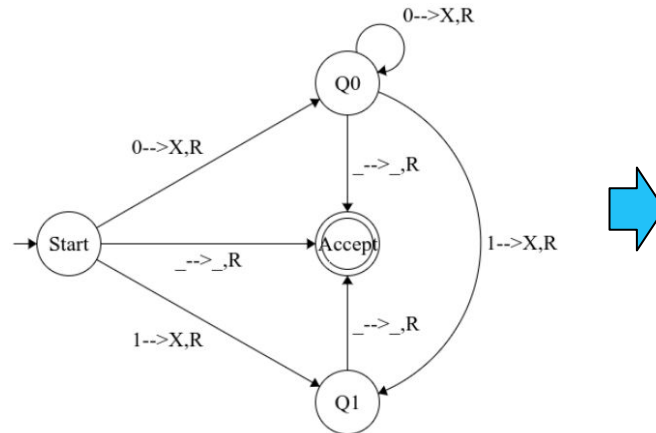
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



If input string is :
00...0111..11
Self-Loop is added



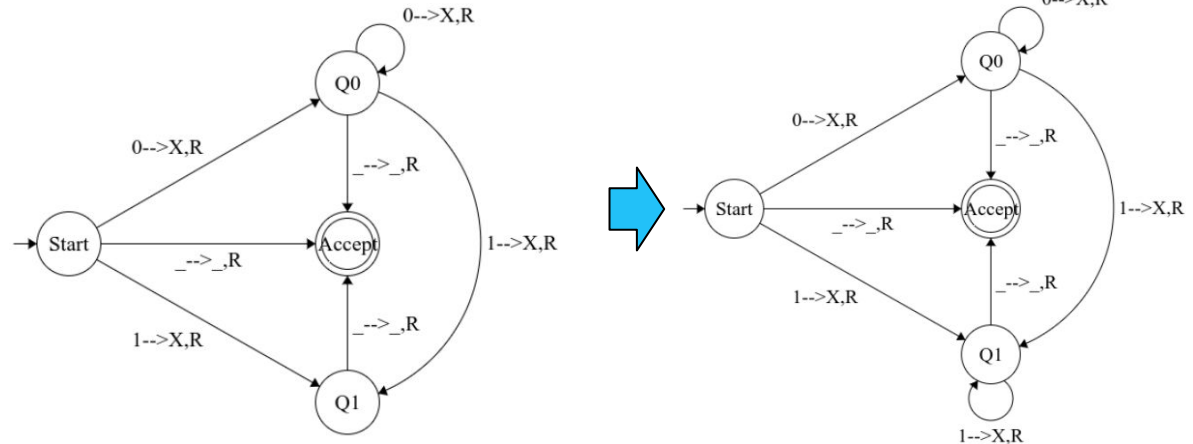
Examples for TM :

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- For the finite automaton of the Turing Machine :



If input string is :
00...0111..11
Self-Loop is added



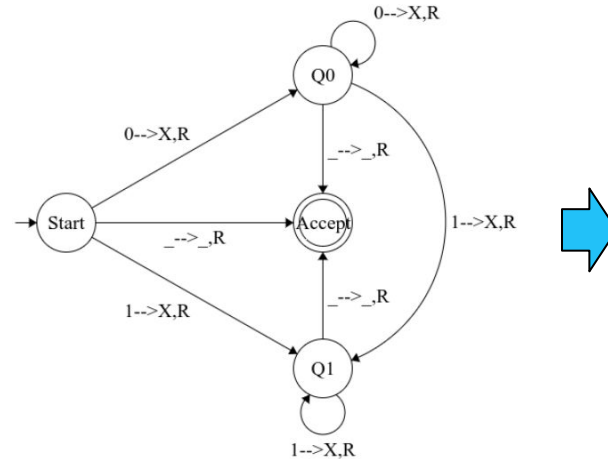
Examples for TM :

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- For the finite automaton of the Turing Machine :



If input string is :
00...010011..11



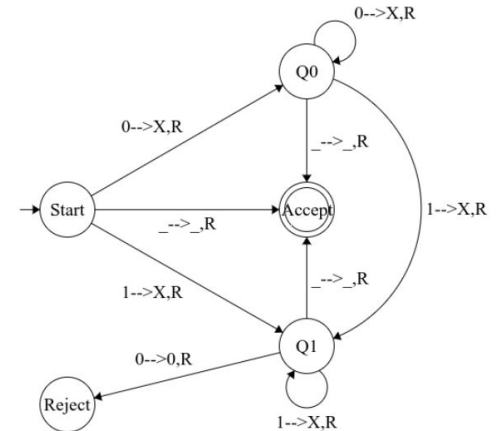
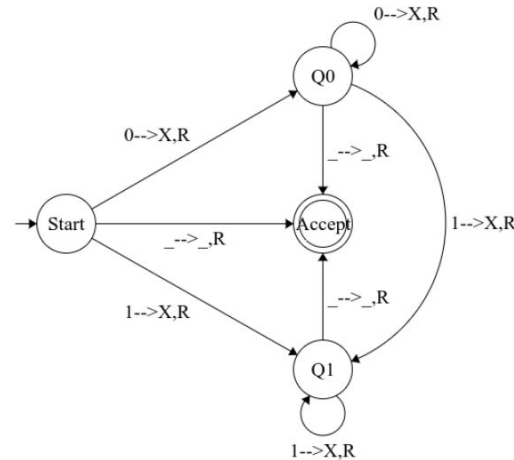
Examples for TM :

0^*1^*

- For the finite automaton of the Turing Machine :



If input string is :
00...010011..11



Examples for TM :

0^*1^*

Online Simulator : <https://turingmachine.io/>

input: '000011111'

blank: ''

start state: start

table:

start:

0: {write: X, R: Q0}

1: {write: Y, R: Q1}

': {R: Accept}

Q0:

0: {write: X, R: Q0}

1: {write: Y, R: Q1}

': {R: Accept}

Q1:

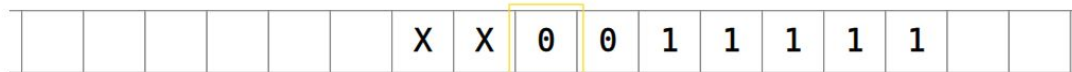
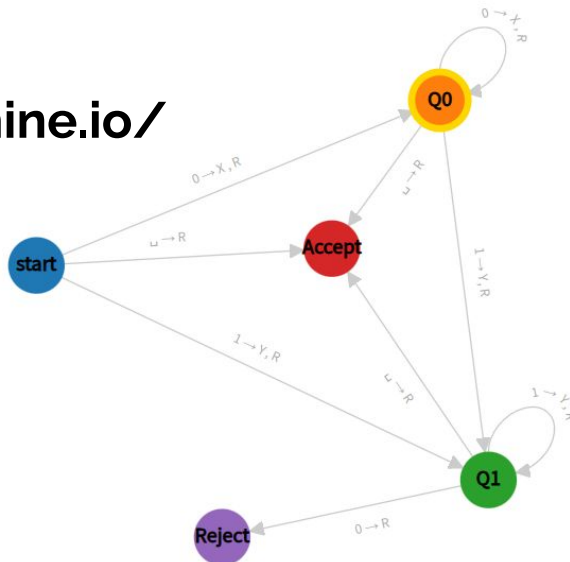
1: {write: Y, R: Q1}

0: {R: Reject}

': {R: Accept}

Accept:

Reject:



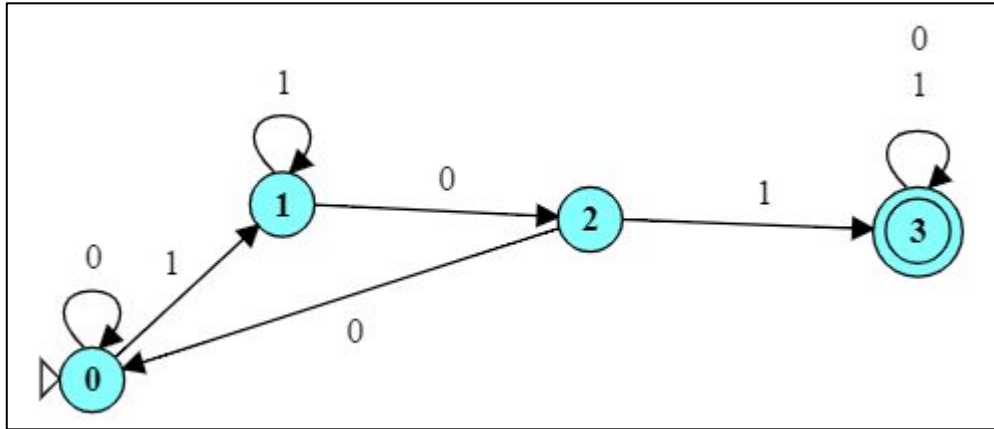
Examples for TM : Contains 101

- The Deterministic Finite Automaton for the language:

Examples for TM :

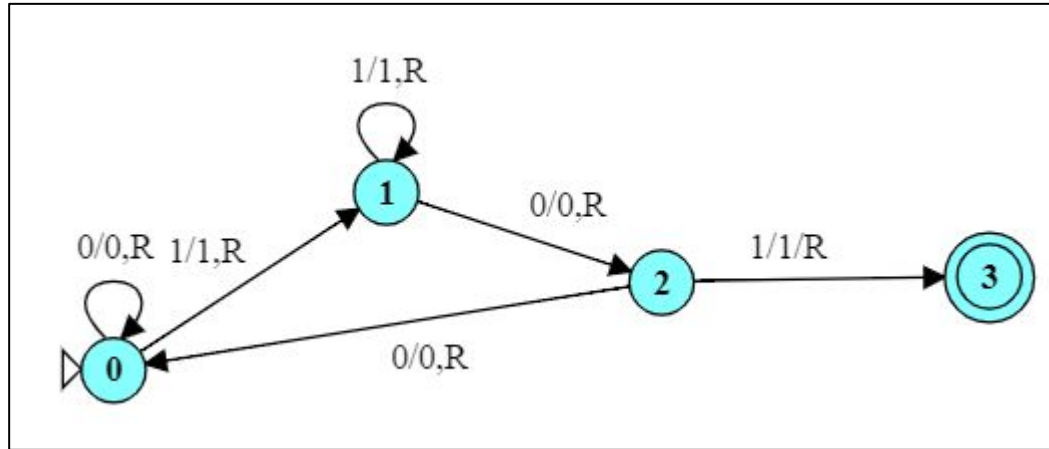
Contains 101

- The Deterministic Finite Automaton for the language:



Examples for TM : Contains 101

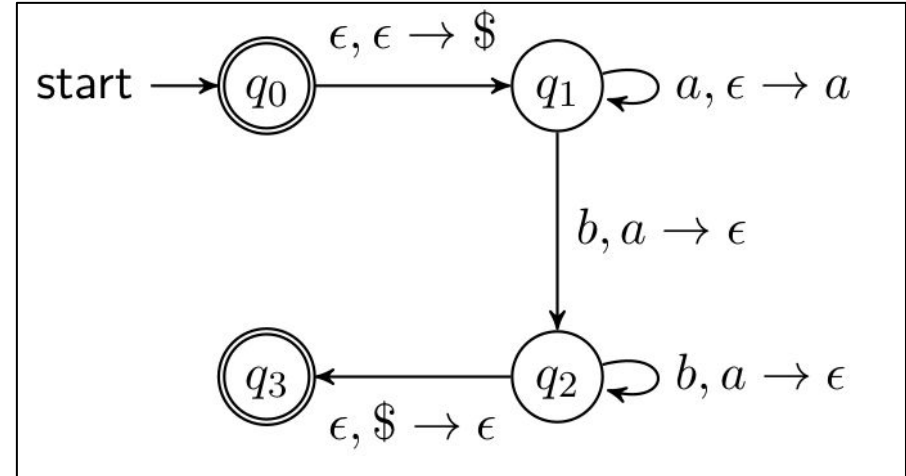
- The Turing Machine for the language is given as:



Examples for TM :

$0^n 1^n$

- The Pushdown Automaton for the language is given as
 - PDA uses a Stack.



Examples for TM :

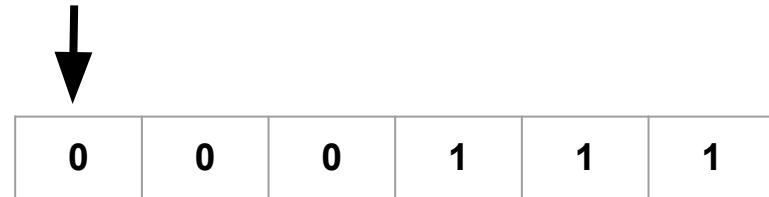
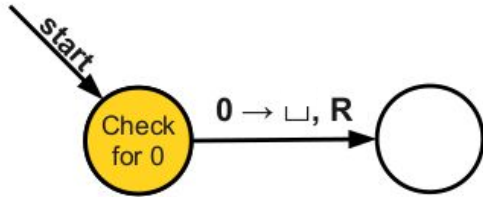
$0^n 1^n$

- The Algorithm for using Turing Machine :
 - ?
- Some basic rules:
 - The string ϵ is in L.
 - Any string starting with 1 is not in L.
 - Any string ending with 0 is not in L.

Examples for TM :

$0^n 1^n$

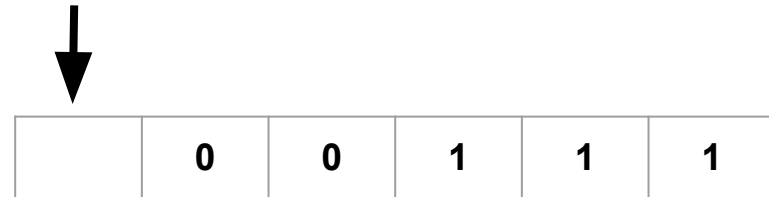
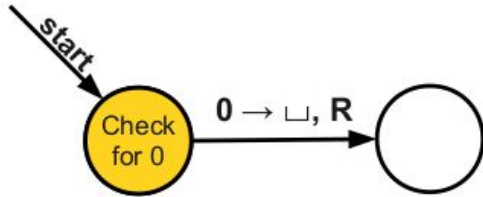
- The Algorithm for using Turing Machine :
 - The **initial** zero found, we change it to blank



Examples for TM :

$0^n 1^n$

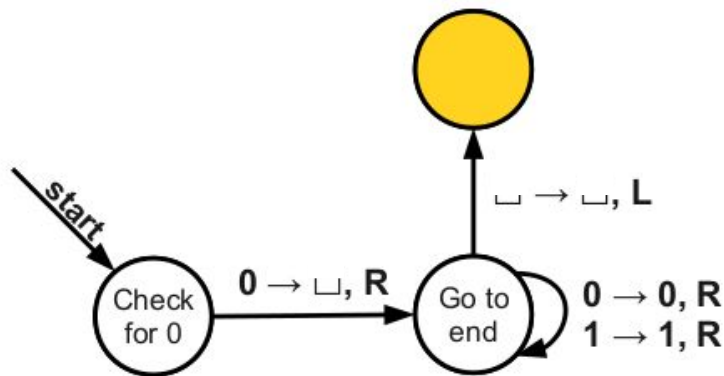
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Examples for TM :

$0^n 1^n$

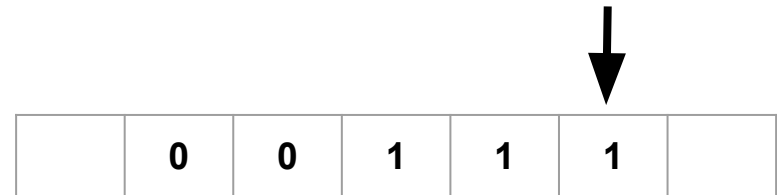
- The Algorithm for using Turing Machine :
 - We search by **skipping all zeros and ones until we reach blank at the extreme right**



Examples for TM :

$0^n 1^n$

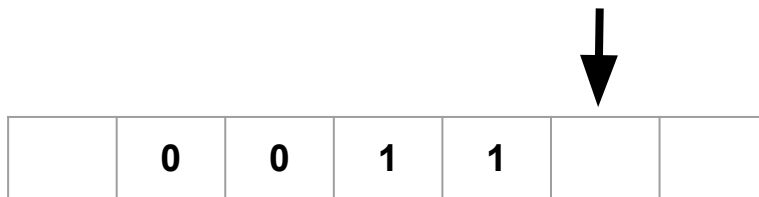
- The Algorithm for using Turing Machine :
 - We search by **skipping all zeros and ones until we reach blank at the extreme right**
 - We move left to the one



Examples for TM :

$0^n 1^n$

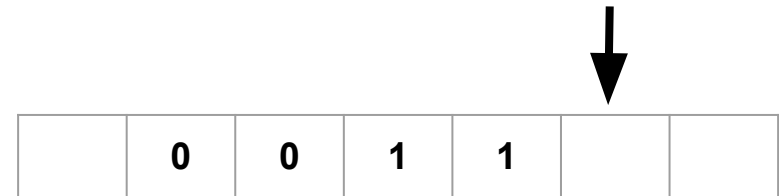
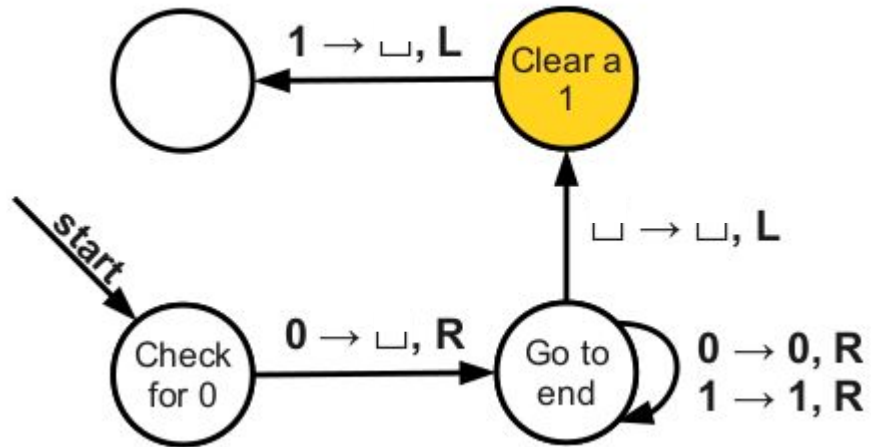
- The Algorithm for using Turing Machine :
 - We search by **skipping all zeros and ones until we reach blank at the extreme right**
 - We move left to the one
 - We replace it with Blank



Examples for TM :

$0^n 1^n$

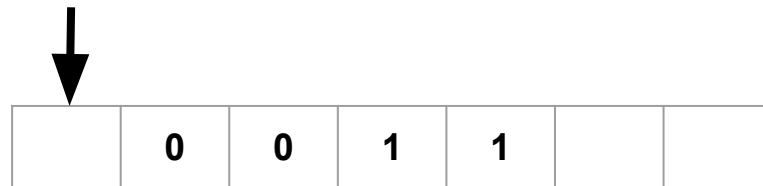
- The Algorithm for using Turing Machine :



Examples for TM :

$0^n 1^n$

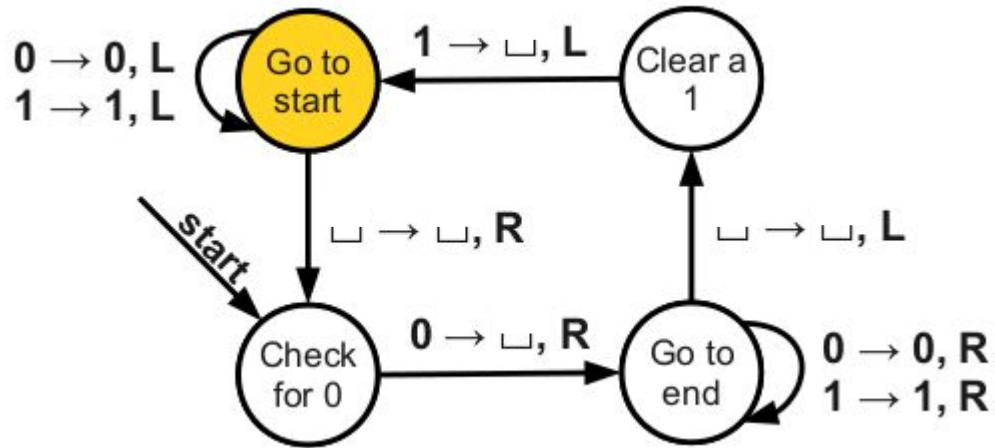
- The Algorithm for using Turing Machine :
 - We search by **skipping all zeros and ones until we reach blank at the extreme LEFT** where you can move right



Examples for TM :

$0^n 1^n$

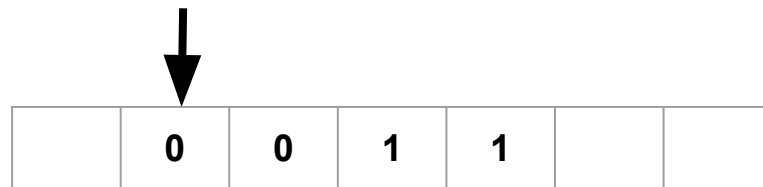
- The Algorithm for using Turing Machine



Examples for TM :

$0^n 1^n$

- The Algorithm for using Turing Machine :
 - We search by **skipping all zeros and ones until we reach blank at the extreme LEFT** where you can move right
 - If at current cell with zero, recursively....



Examples for TM :

$0^n 1^n$

- The Algorithm for using Turing Machine :
 - When to Stop ? Let's assume for an Accept:



Examples for TM :

$0^n 1^n$

- The Algorithm for using Turing Machine :
 - When to Stop ? Let's assume for an Accept:
 - When there is a blank during the start state



Examples for TM :

$0^n 1^n$

- The Algorithm for using Turing Machine :
 - When to say whether a word is accepted ?
 - By DFA/NFA
 - By PDA
 - By TM

Examples for TM :

$0^n 1^n$

- The Algorithm for using Turing Machine :
 - When to say whether a word is accepted ?
 - By DFA/NFA
 - By PDA
 - By TM

1. You reach an accept state
2. You read all letters in the tape (given string)

Examples for TM :

$0^n 1^n$

- The Algorithm for using Turing Machine :
 - When to say whether a word is accepted ?

- By DFA/NFA

Strictly

- By PDA

1. You reach an accept state
2. You read all letters in the tape (given string)

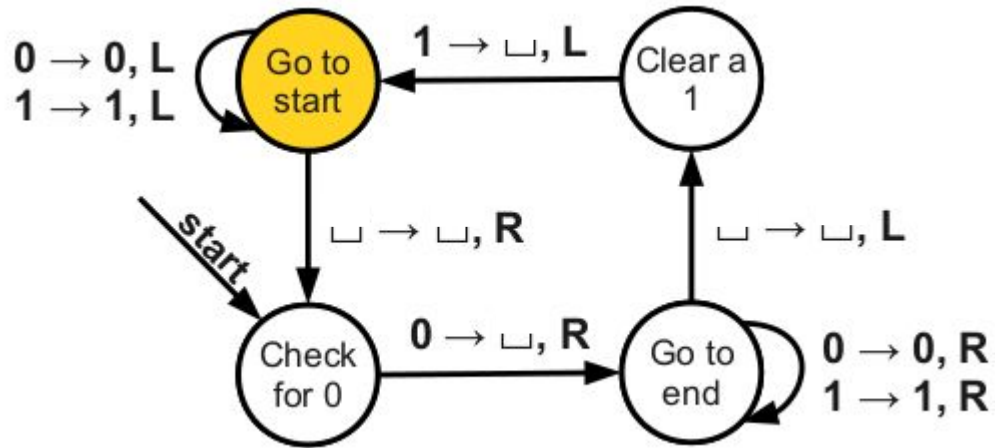
- By TM

1. Strictly, you reach an accept state
2. Optionally read all letters ,,,(depending on the language, TM...

Examples for TM :

$0^n 1^n$

- The Algorithm for using Turing Machine



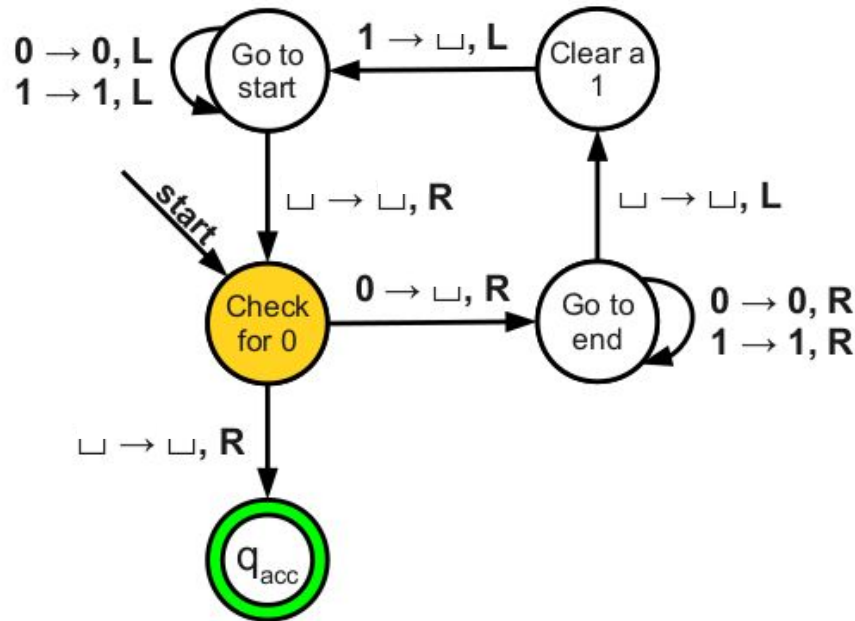
How to stop and accept ?



Examples for TM :

$0^n 1^n$

- The Algorithm for using Turing Machine :



Examples for TM :

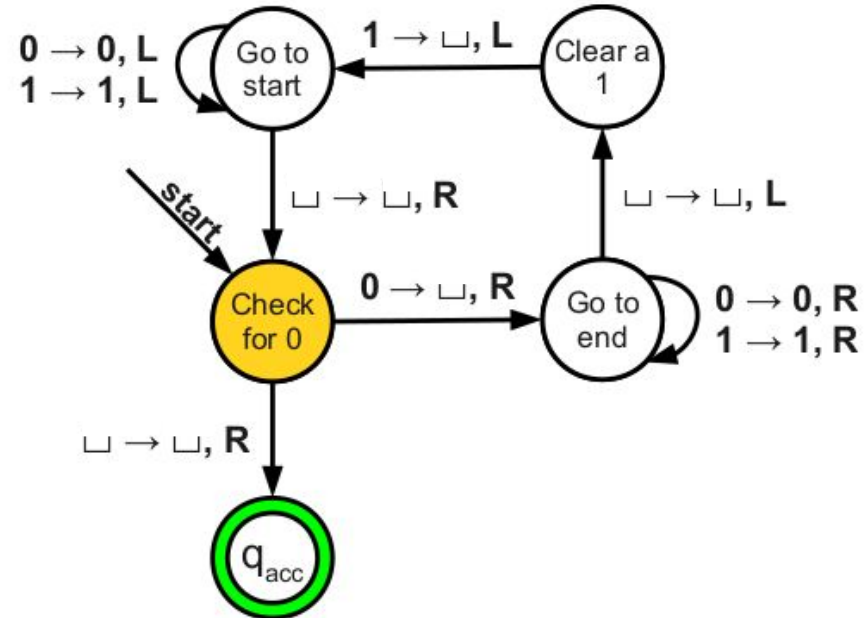
$0^n 1^n$

- The Algorithm for using Turing Machine :

- Following words:

- 1
- 01111
- 001

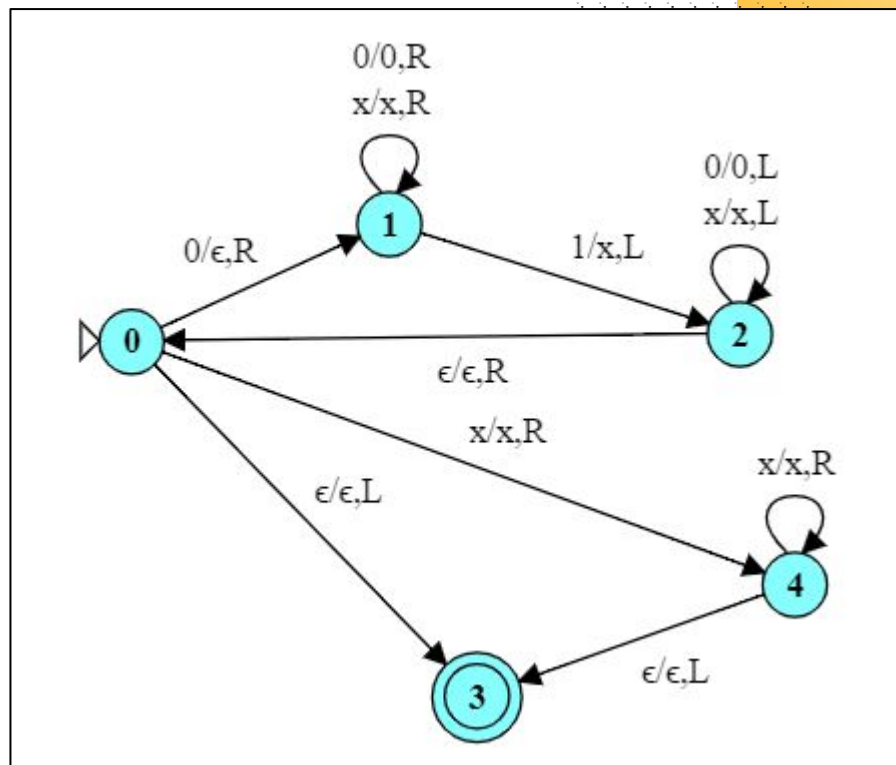
- Reject state and transitions are implicit



Examples for TM :

$0^n 1^n$

- Another Possible solution



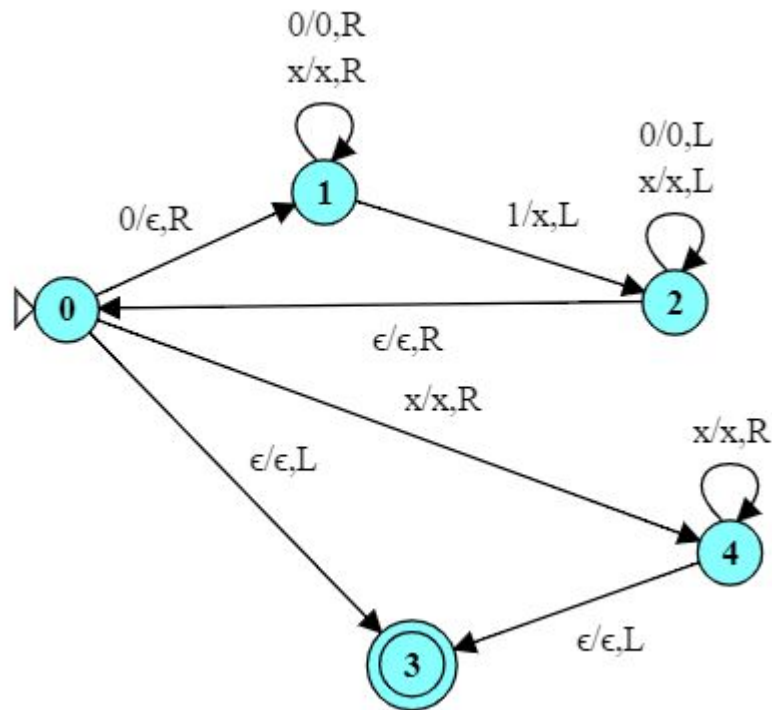
Examples for TM :

$0^n 1^n$

- Another Possible solution

Don't be confused with the
Epsilon of PDA/DFA

Here means the tape cell is
blank...

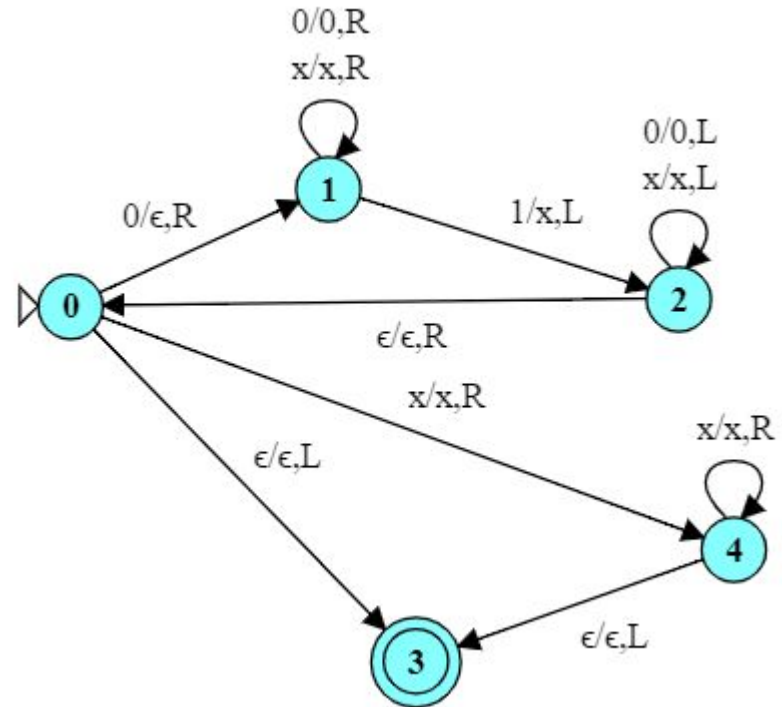


Examples for TM :

$0^n 1^n$

- Another Possible solution

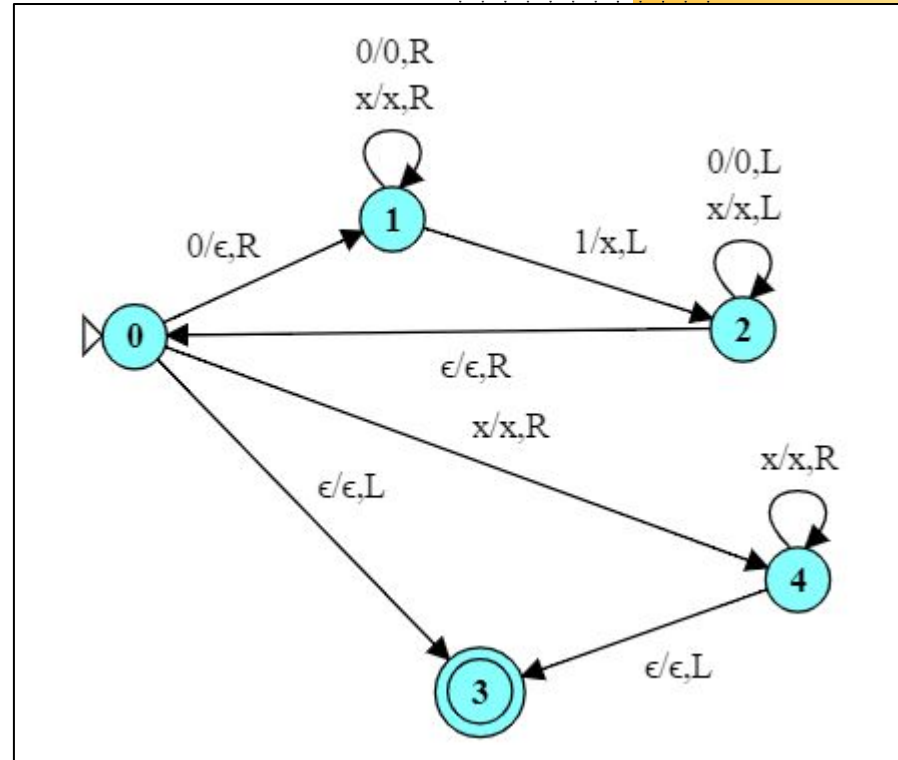
Deduce the Algorithm being used



Examples for TM :

$0^n 1^n$

- Is the following word accepted :
 - **01111**
- Notations:
 - $0/x,R$: when you read 0, replace it with X and move Right.

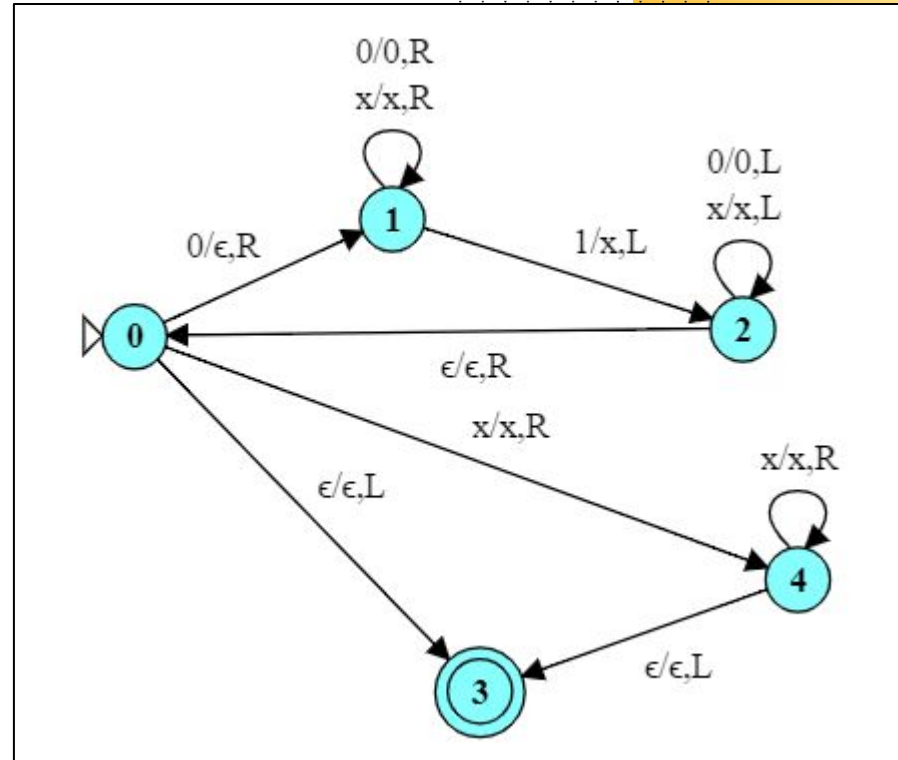


Examples for TM :

$0^n 1^n$

- Is the following word accepted : **0011**

Step	State	0	0	1	1
0	0	0	0	1	1

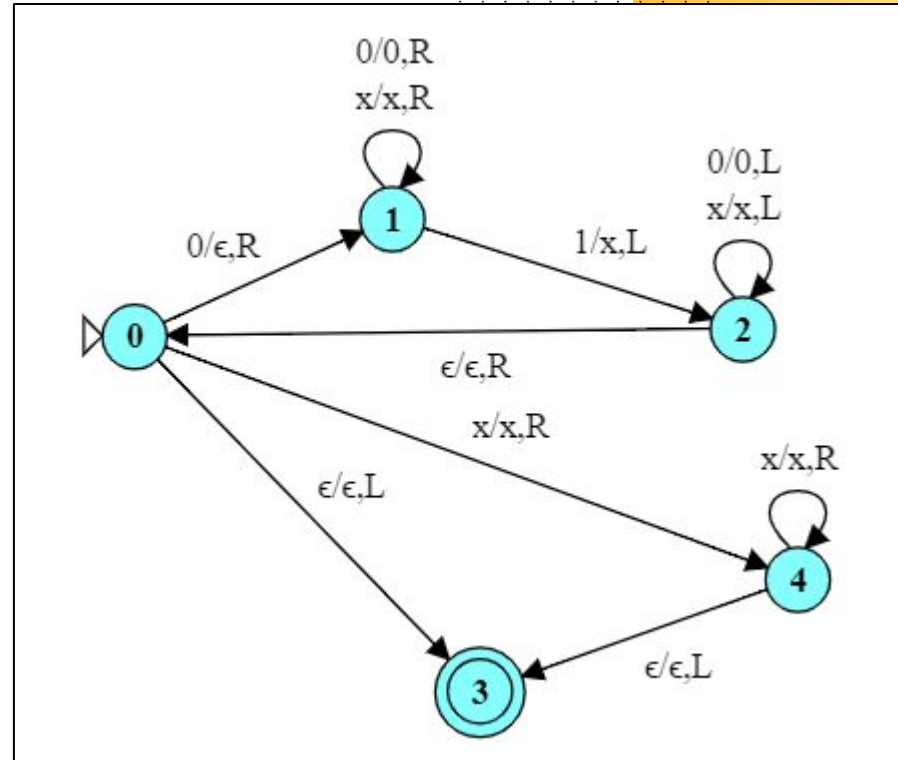


Examples for TM :

$0^n 1^n$

- Is the following word accepted : **0011**

Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1

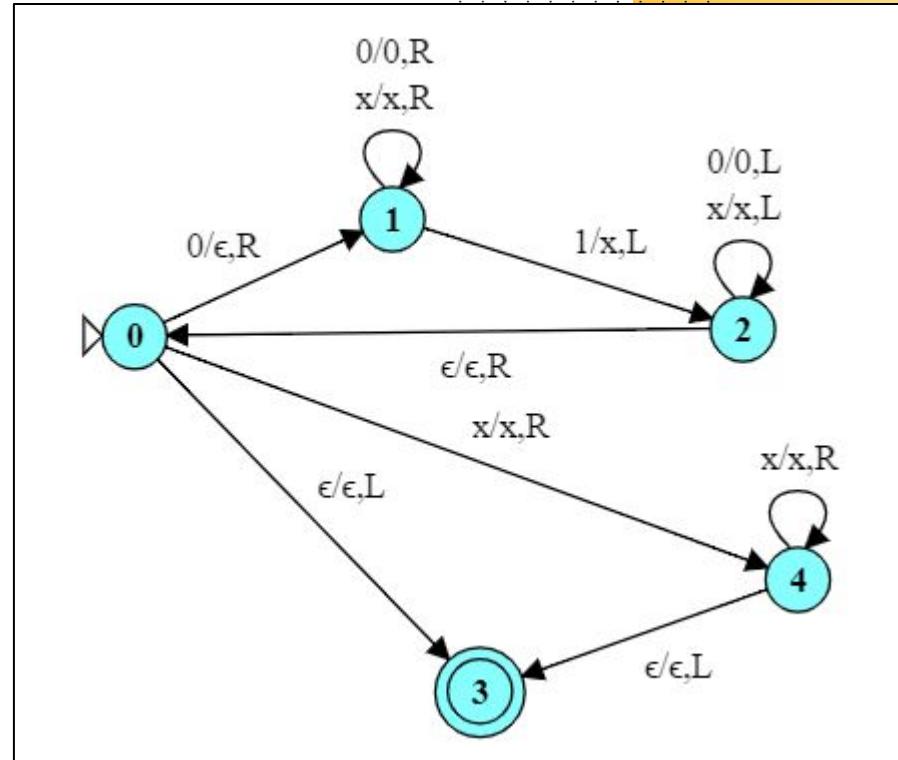


Examples for TM :

$0^n 1^n$

- Is the following word accepted : **0011**

Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1

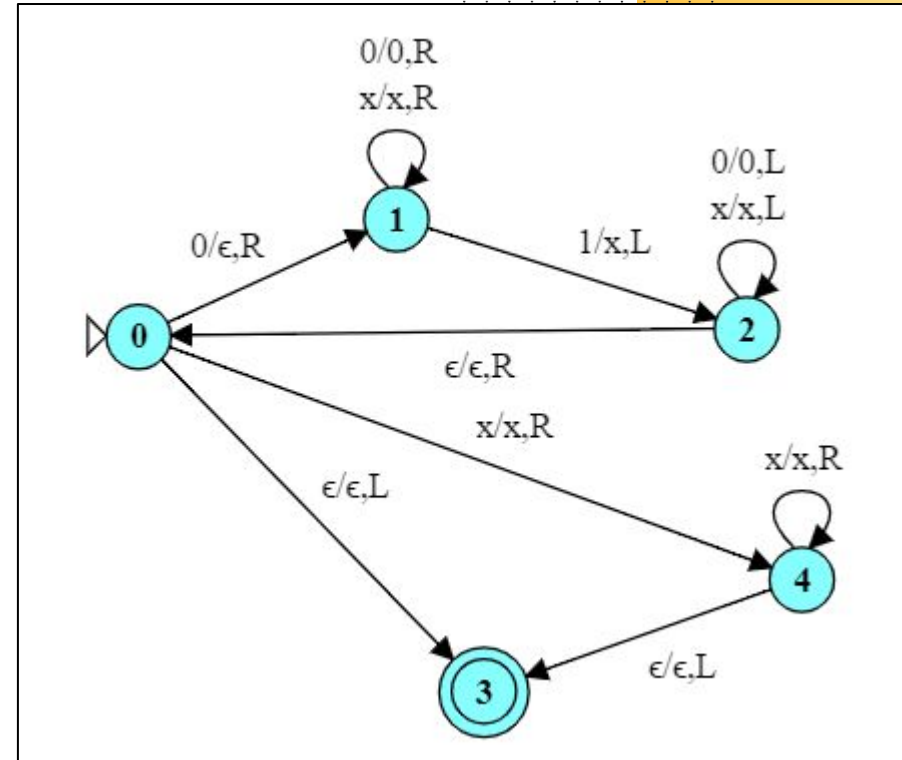


Examples for TM :

$0^n 1^n$

- Is the following word accepted : **0011**

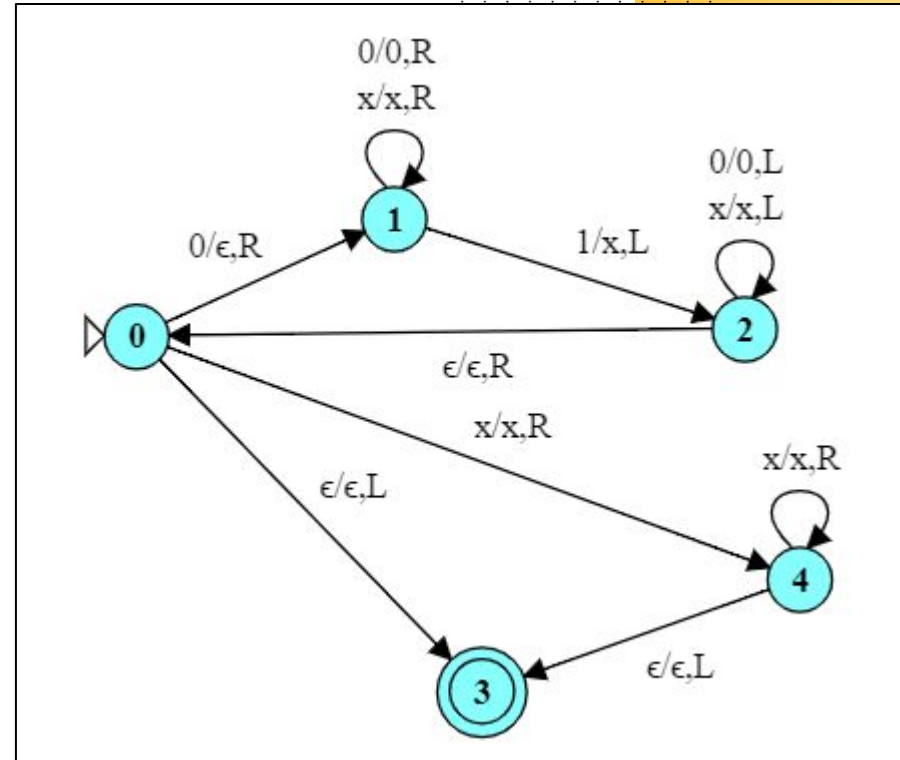
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1
3	2	□	0	x	1



Examples for TM :

$0^n 1^n$

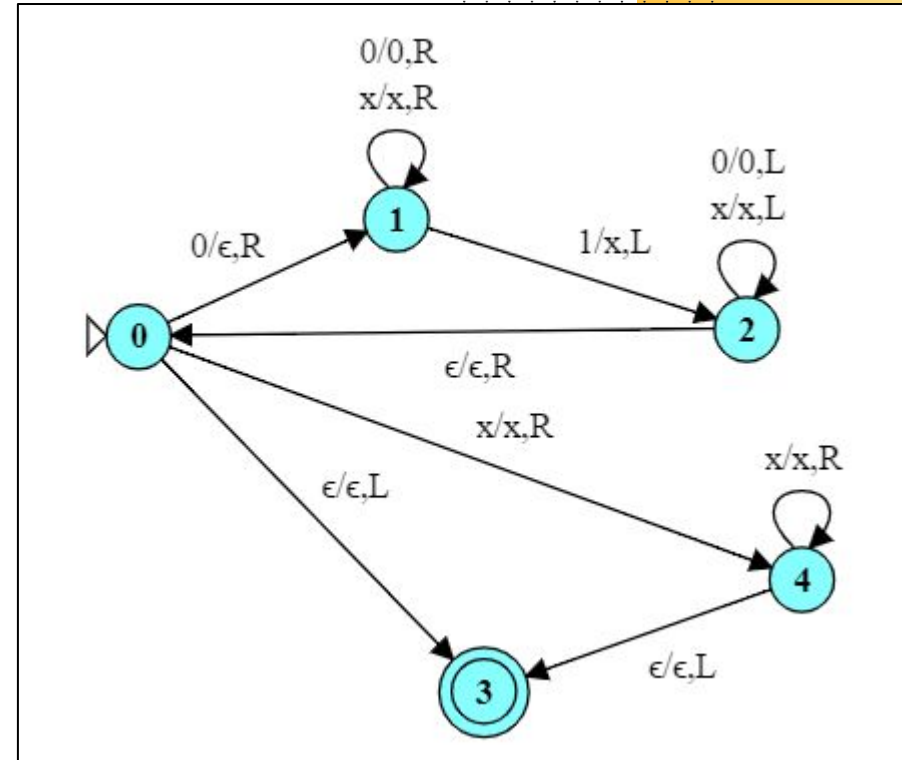
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1
3	2	□	0	x	1
4	2	□	0	x	1



Examples for TM :

$0^n 1^n$

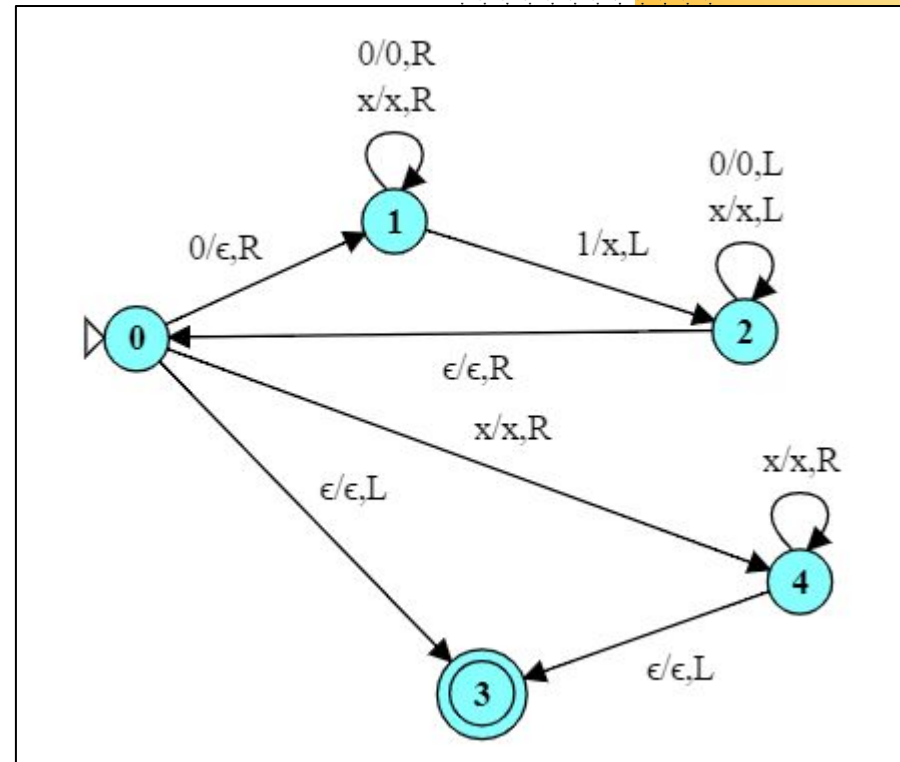
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1
3	2	□	0	x	1
4	2	□	0	x	1
5	0	□	0	x	1



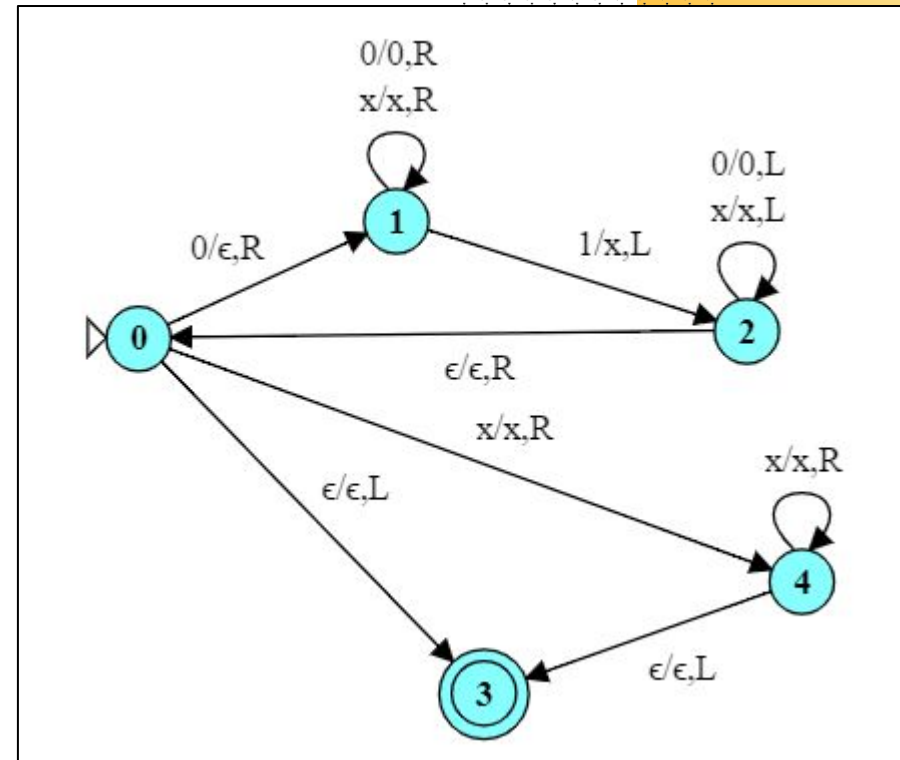
Examples for TM :

$0^n 1^n$

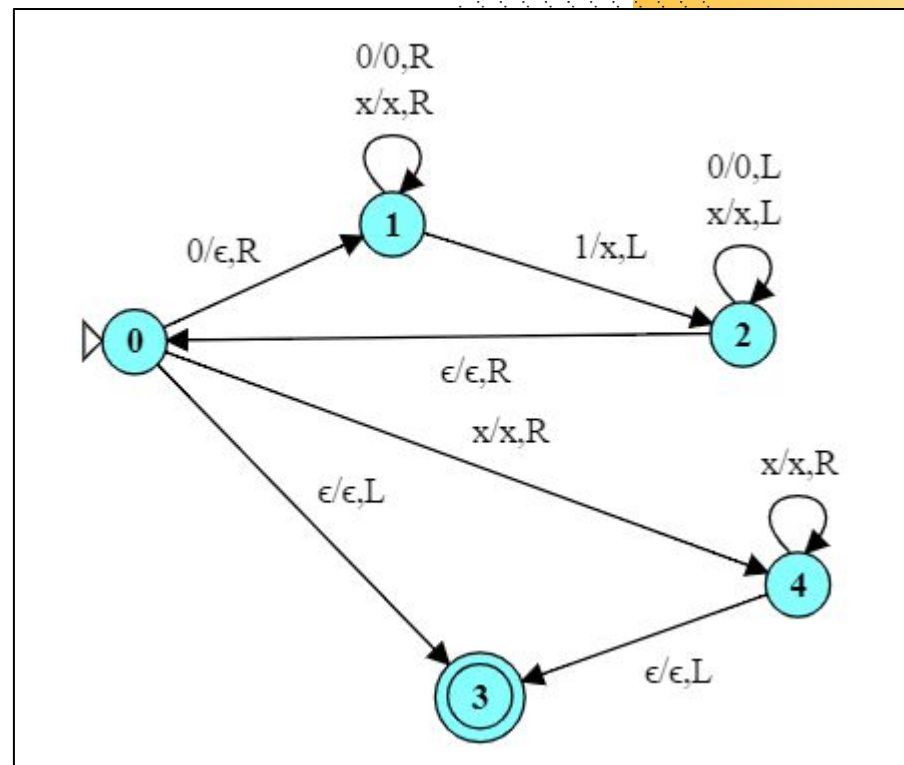
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1
3	2	□	0	x	1
4	2	□	0	x	1
5	0	□	0	x	1
6	1	□	□	x	1



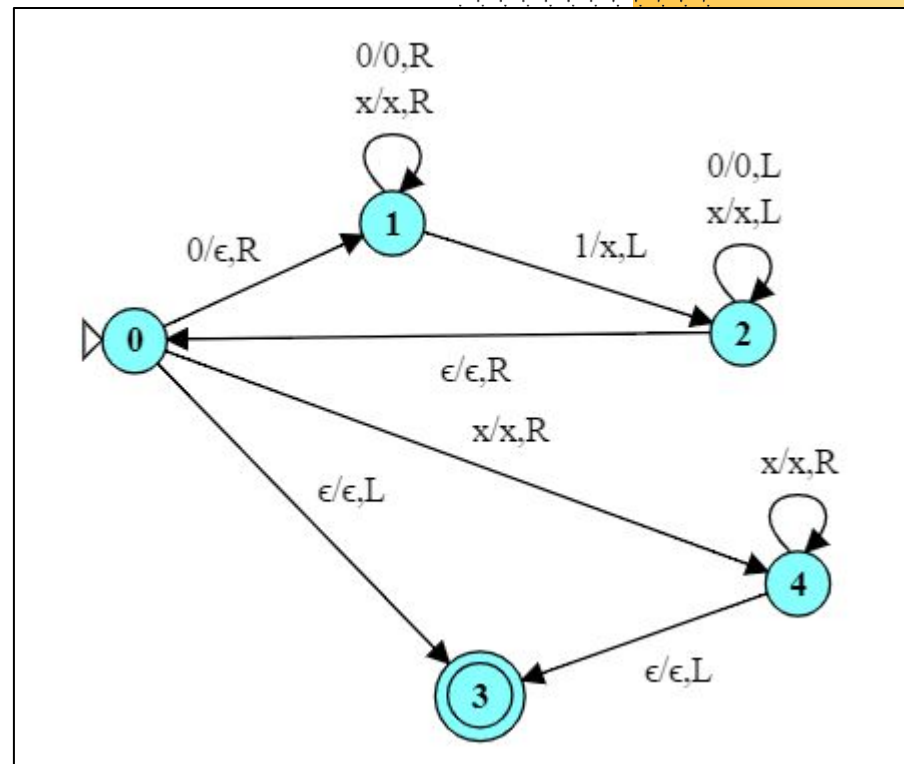
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1
3	2	□	0	x	1
4	2	□	0	x	1
5	0	□	0	x	1
6	1	□	□	x	1
7	1	□	□	x	1



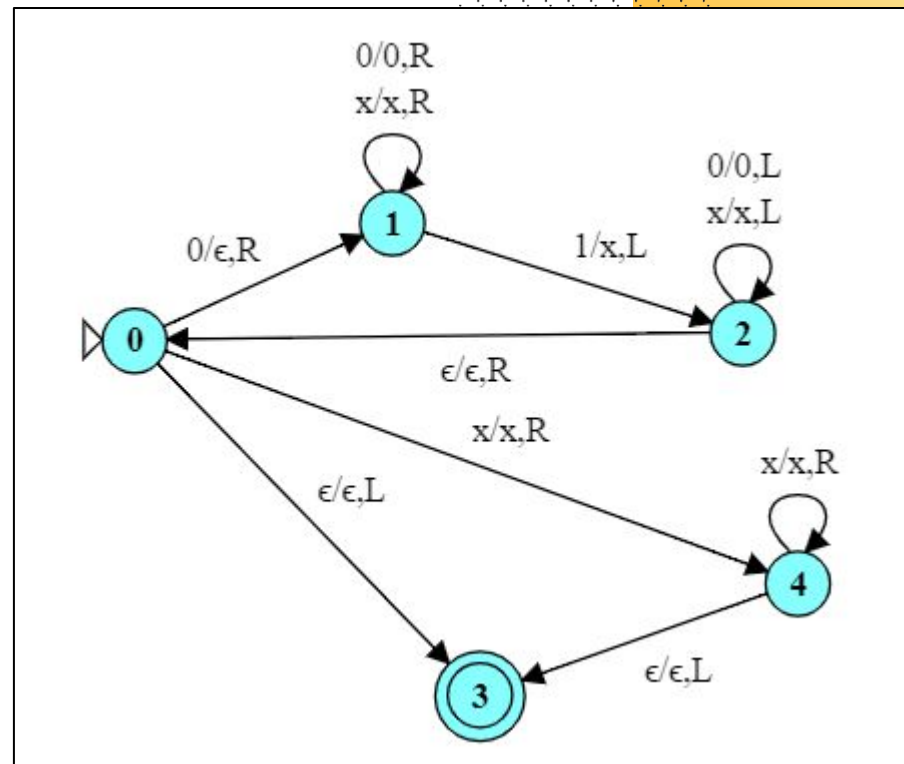
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1
3	2	□	0	x	1
4	2	□	0	x	1
5	0	□	0	x	1
6	1	□	□	x	1
7	1	□	□	x	1
8	2	□	□	x	x



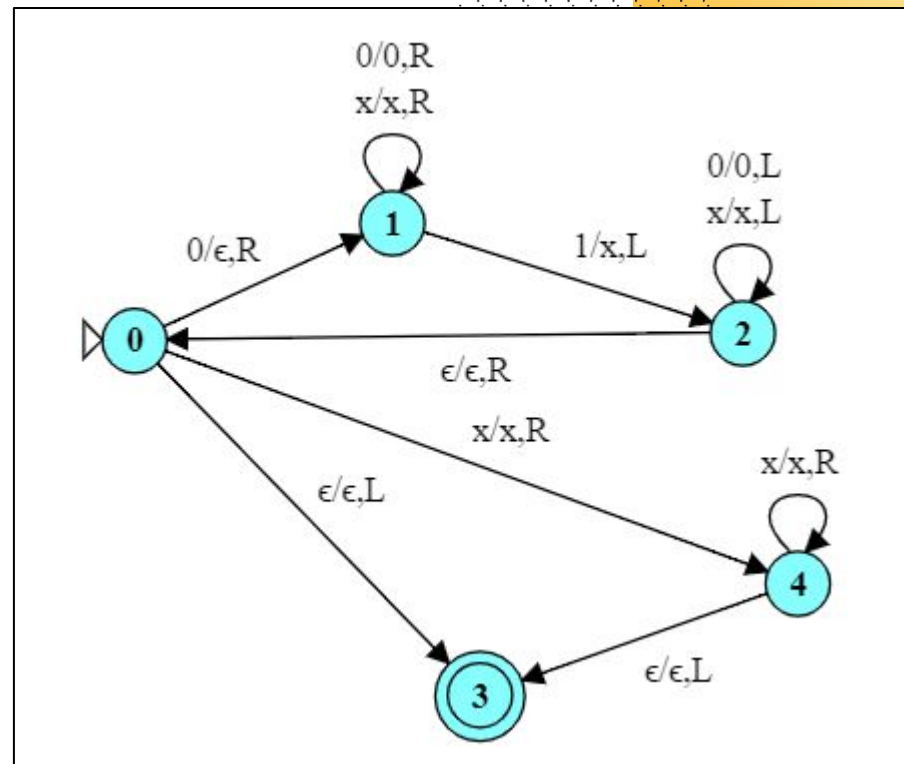
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1
3	2	□	0	x	1
4	2	□	0	x	1
5	0	□	0	x	1
6	1	□	□	x	1
7	1	□	□	x	1
8	2	□	□	x	x
9	2	□	□	x	x



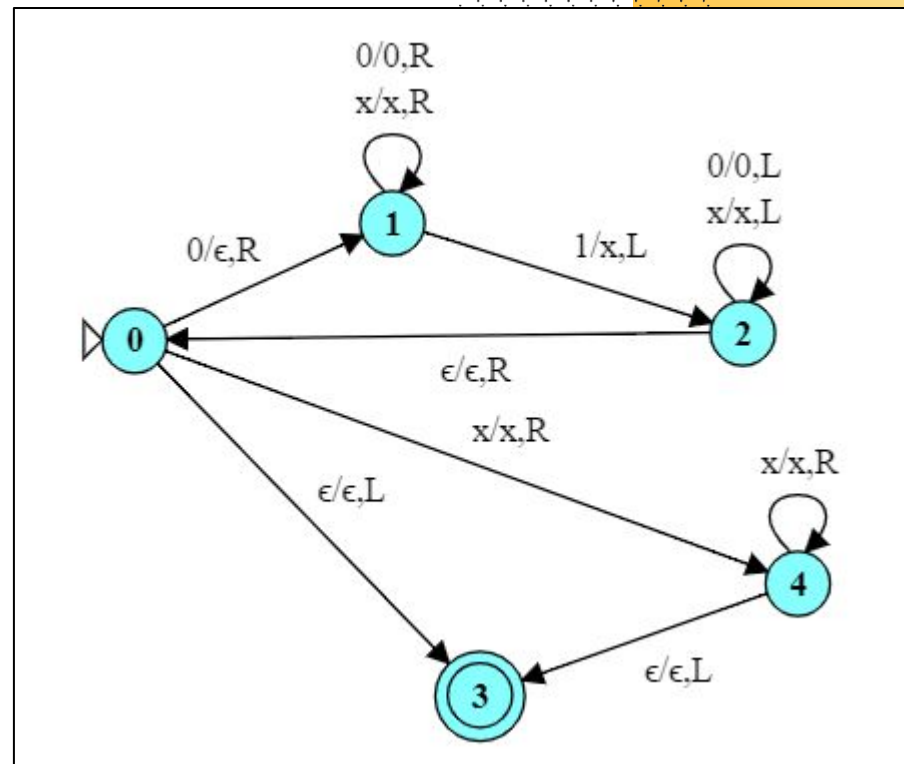
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	□	0	1	1
2	1	□	0	1	1
3	2	□	0	x	1
4	2	□	0	x	1
5	0	□	0	x	1
6	1	□	□	x	1
7	1	□	□	x	1
8	2	□	□	x	x
9	2	□	□	x	x
10	0	□	□	x	x



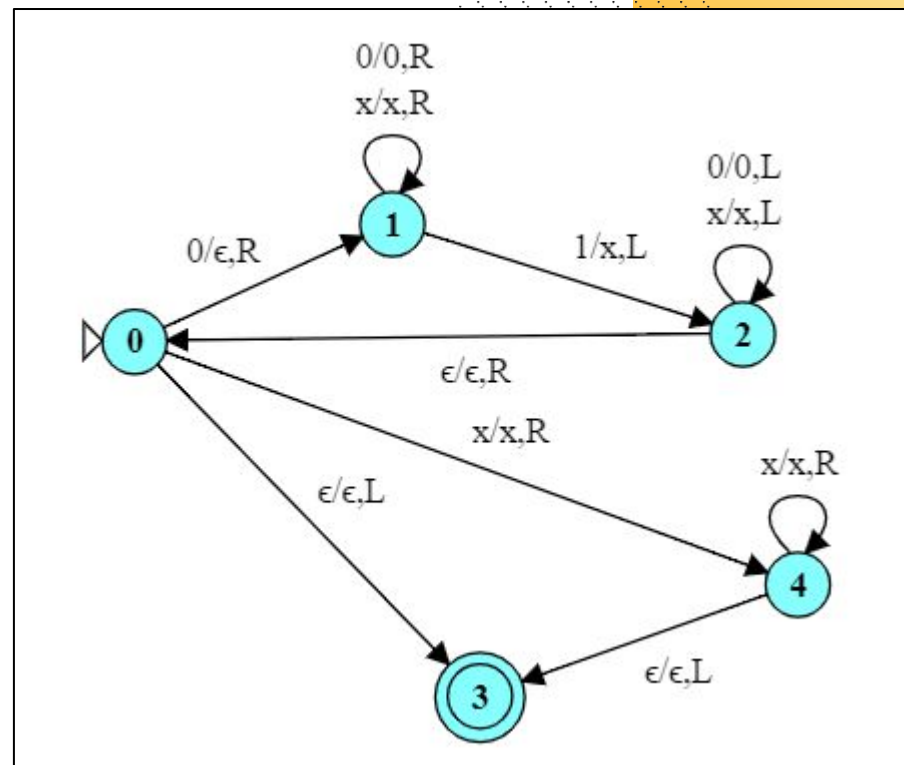
Step	State	0	0	1	1
0	0	0	0	1	1
1	1	␣	0	1	1
2	1	␣	0	1	1
3	2	␣	0	x	1
4	2	␣	0	x	1
5	0	␣	0	x	1
6	1	␣	␣	x	1
7	1	␣	␣	x	1
8	2	␣	␣	x	x
9	2	␣	␣	x	x
10	0	␣	␣	x	x
11	4	␣	␣	x	x



Step	State	0	0	1	1	
0	0	0	0	1	1	
1	1	□	0	1	1	
2	1	□	0	1	1	
3	2	□	0	x	1	
4	2	□	0	x	1	
5	0	□	0	x	1	
6	1	□	□	x	1	
7	1	□	□	x	1	
8	2	□	□	x	x	
9	2	□	□	x	x	
10	0	□	□	x	x	
11	4	□	□	x	x	
12	4	□	□	x	x	



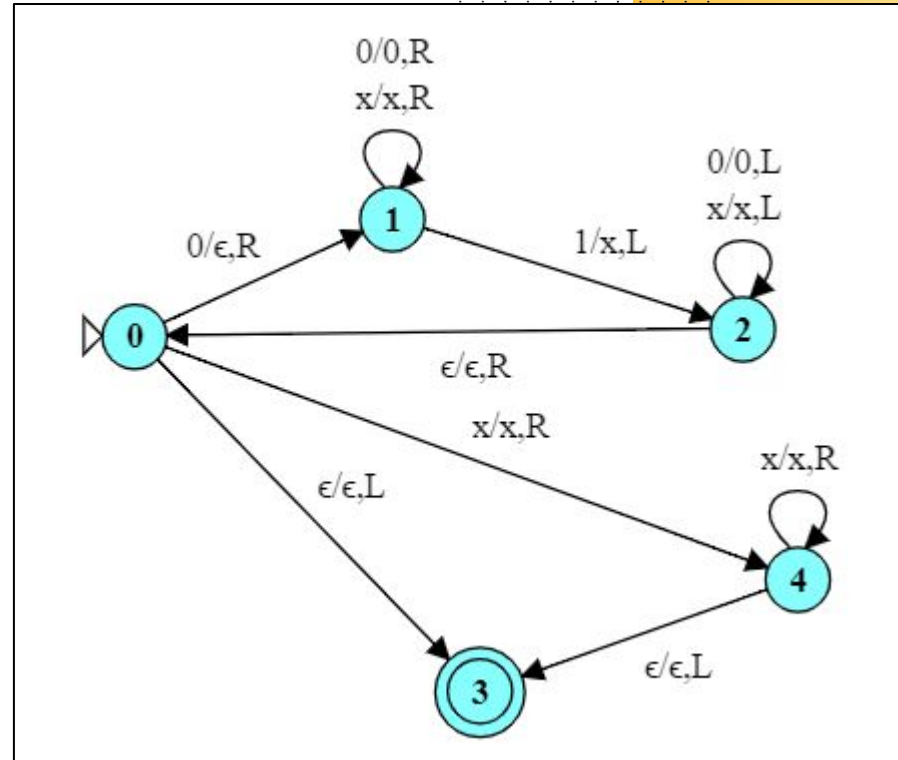
Step	State	0	0	1	1	
0	0	0	0	1	1	
1	1	□	0	1	1	
2	1	□	0	1	1	
3	2	□	0	x	1	
4	2	□	0	x	1	
5	0	□	0	x	1	
6	1	□	□	x	1	
7	1	□	□	x	1	
8	2	□	□	x	x	
9	2	□	□	x	x	
10	0	□	□	x	x	
11	4	□	□	x	x	
12	4	□	□	x	x	
13	3	□	□	x	x	



Examples for TM :

$0^n 1^n$

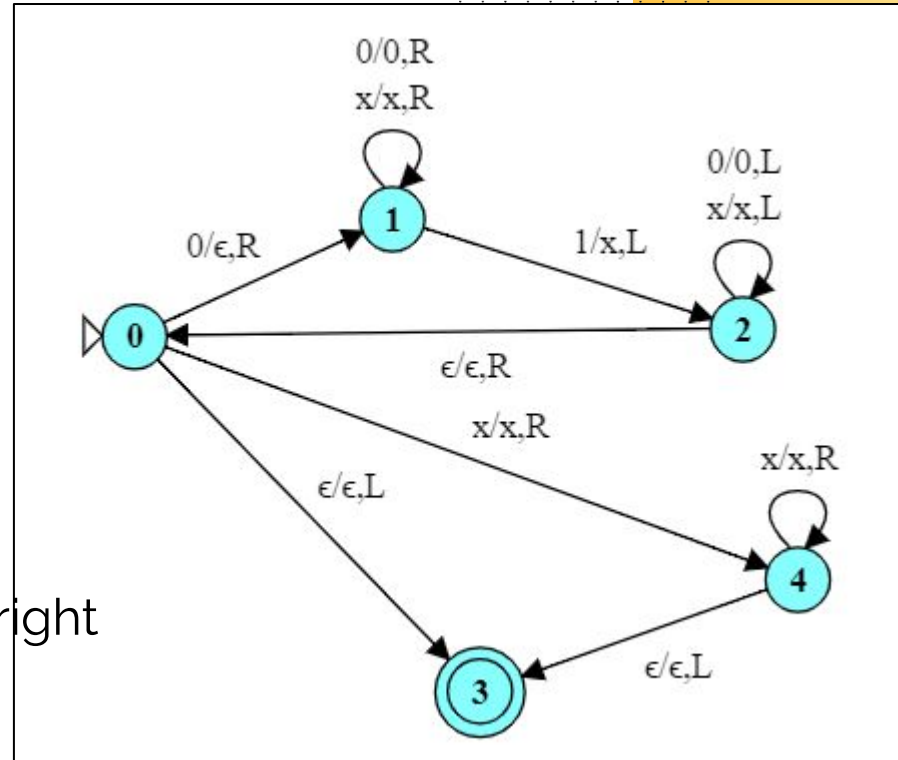
- What's the Algorithm ?



Examples for TM :

$0^n 1^n$

- What's the Algorithm :
 - Initial zero replace by blank
 - Skip right x and 0 till you find first 1
 - Replace 1 by x and move left
 - Skip left 0 and x until blank, move right
 - If no only x or spaces, accept.



Examples for TM :

$a^n b^n c^n$

- This language is :
 - Not Regular, therefore, we cannot create the DFA
 - Not Context Free, therefore we cannot create a Pushdown Automaton
 - But, we can create the Turing Machine for this Language

Examples for TM :

$a^n b^n c^n$

- The Algorithm :



Examples for TM :

$a^n b^n c^n$

- The Algorithm :
 - Replace a with A, skip right all a to find first b.
 - Replace b with B, skip right all b to find first c.
 - Replace c with C, Skip LEFT all a,A,b,B,C until A is found.
 - Keep repeating the process.

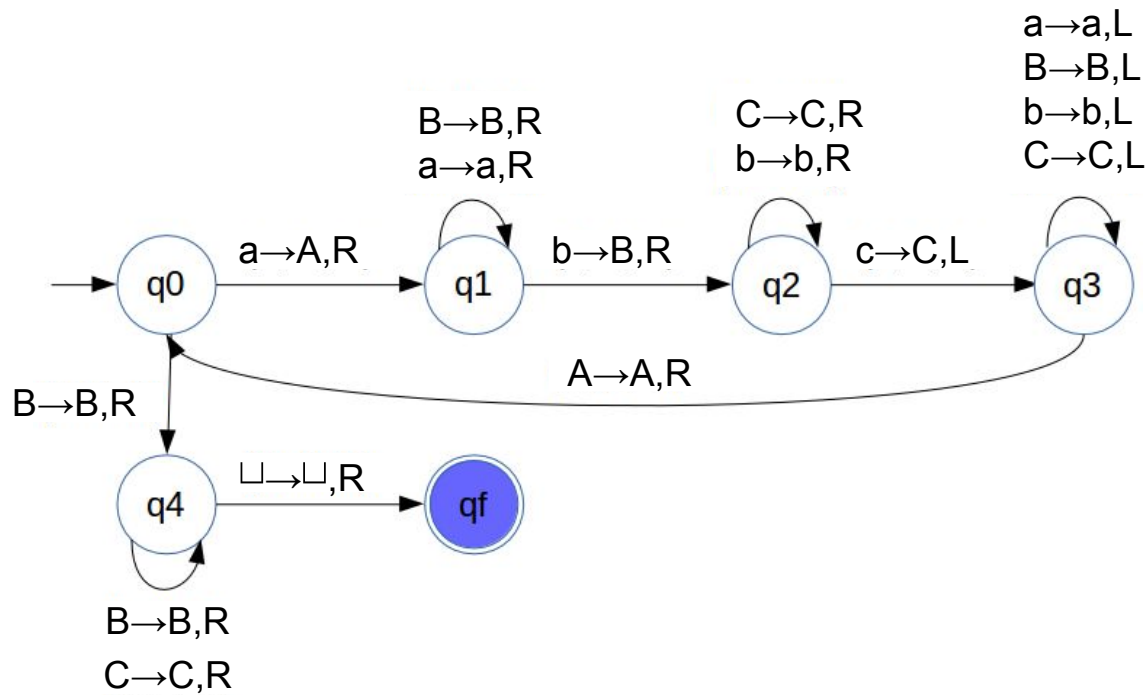
Examples for TM :

$a^n b^n c^n$

- The Algorithm :
 - Replace a with A, skip right all a to find first b.
 - Replace b with B, skip right all b to find first c.
 - Replace c with C, Skip LEFT all a,A,b,B,C until A is found.
 - Keep repeating the process.
 - Until all letters are : A, B, and C on the tape.

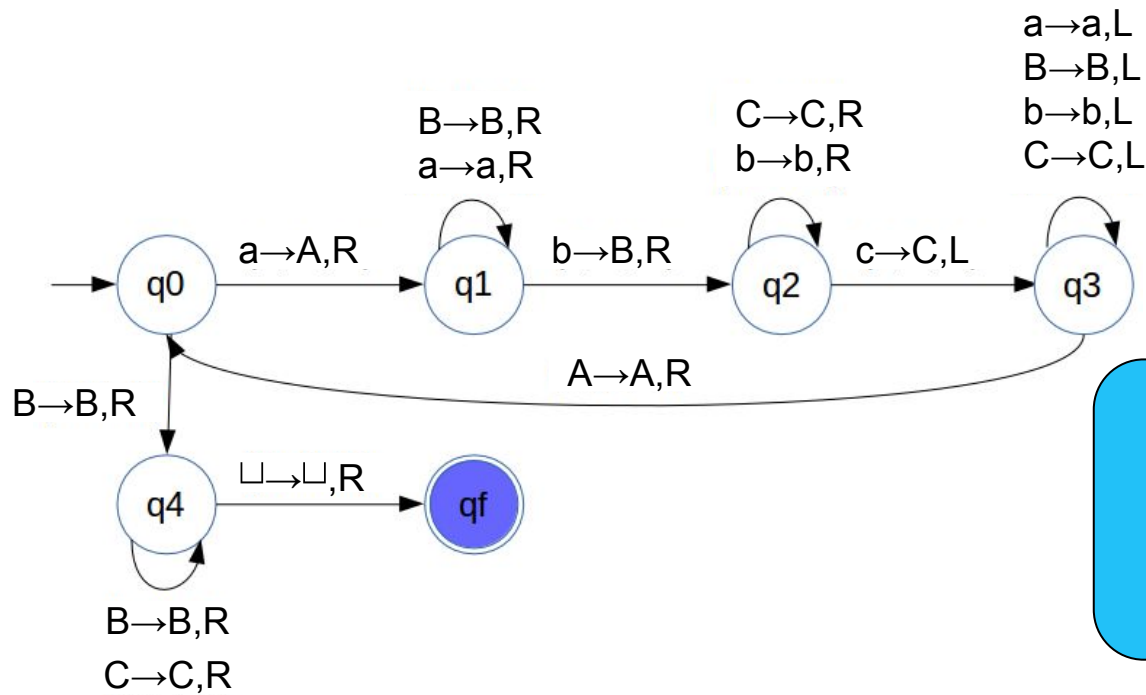
Examples for TM :

$a^n b^n c^n$



Examples for TM :

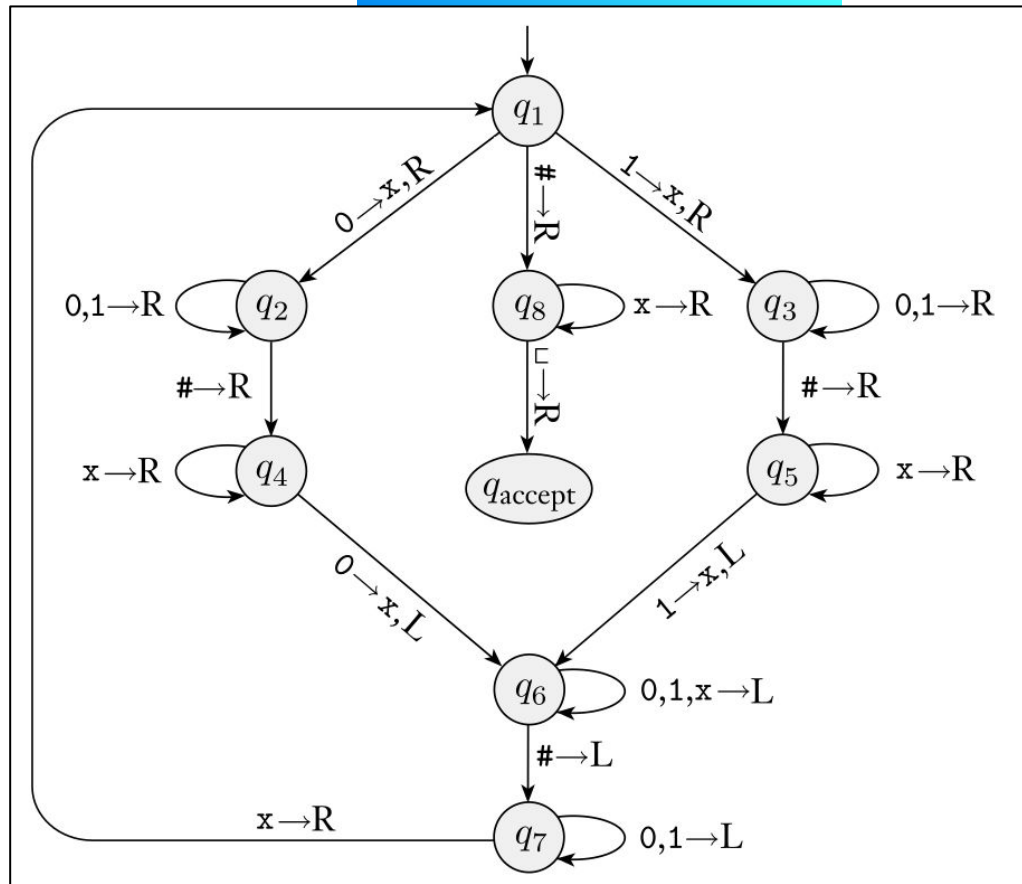
$a^n b^n c^n$



**Is the empty string
accepted ?**

Examples for TM : $w\#w$

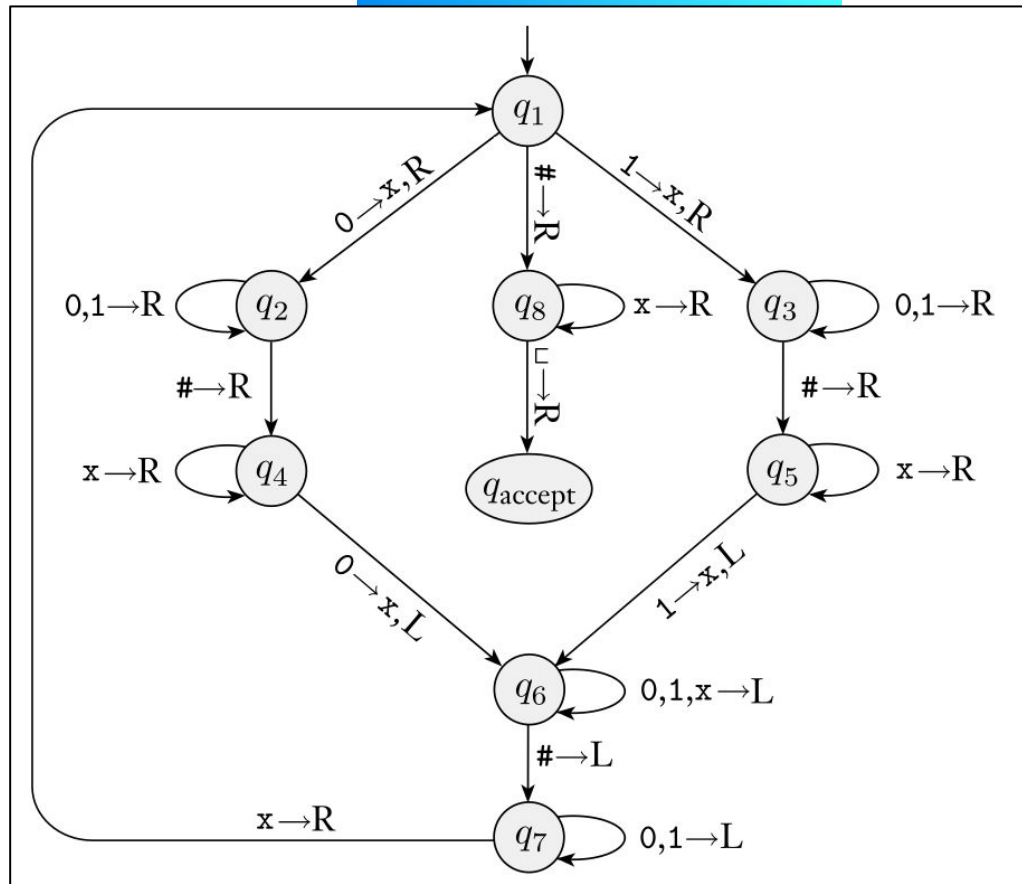
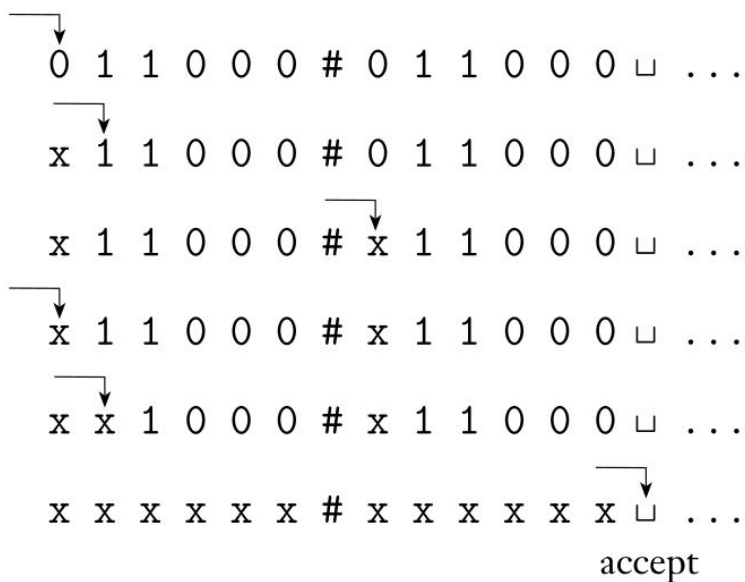
- What's the Algorithm for this ?



Examples for TM :

$w\#w$

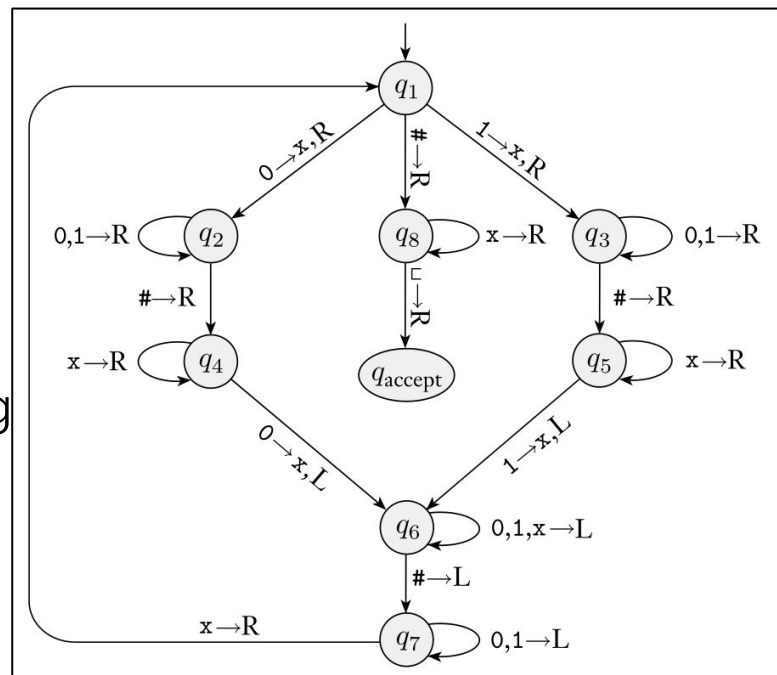
- What's the Algorithm for this ?



Examples for TM :

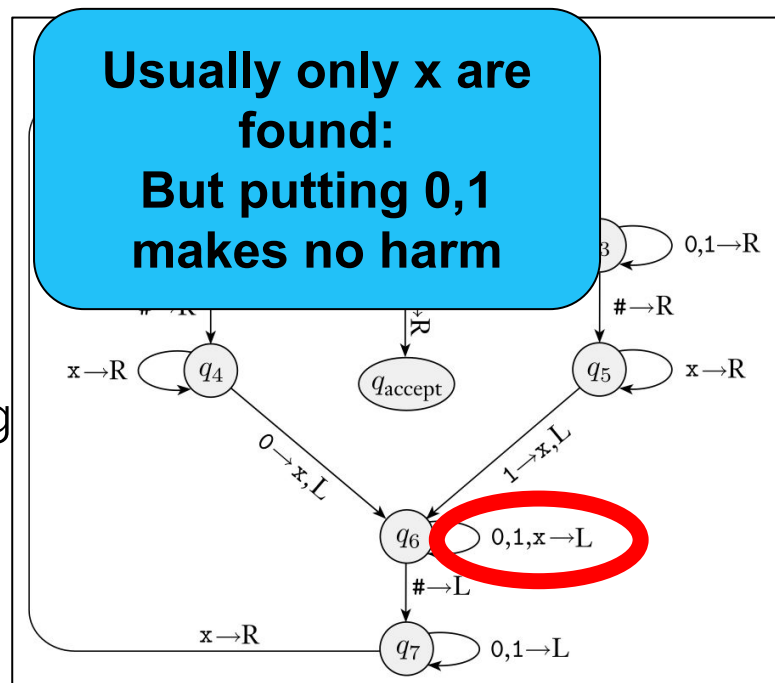
$w\#w$

- What's the Algorithm for this :
 - Initially: For a symbol (1 or 0) $\rightarrow x$, R
 - Skip all 0 and 1 till #
 - Skip **only** x until the same initial symbol
 - Skip **LEFT ONLY** x until #, keep skipping 0 and 1
 - If x is found move right and go to initial



Examples for TM : $w\#w$

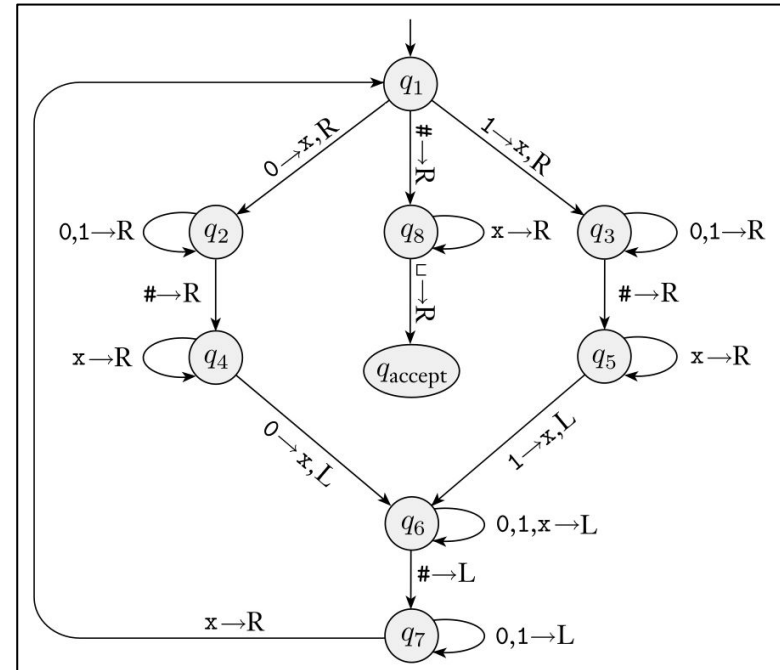
- What's the Algorithm for this :
 - Initially: For a symbol (1 or 0) $\rightarrow x$, R
 - Skip all 0 and 1 till #
 - Skip **only** x until the same initial symbol
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 - If x is found move right and go to initial



Examples for TM :

$w\#w$

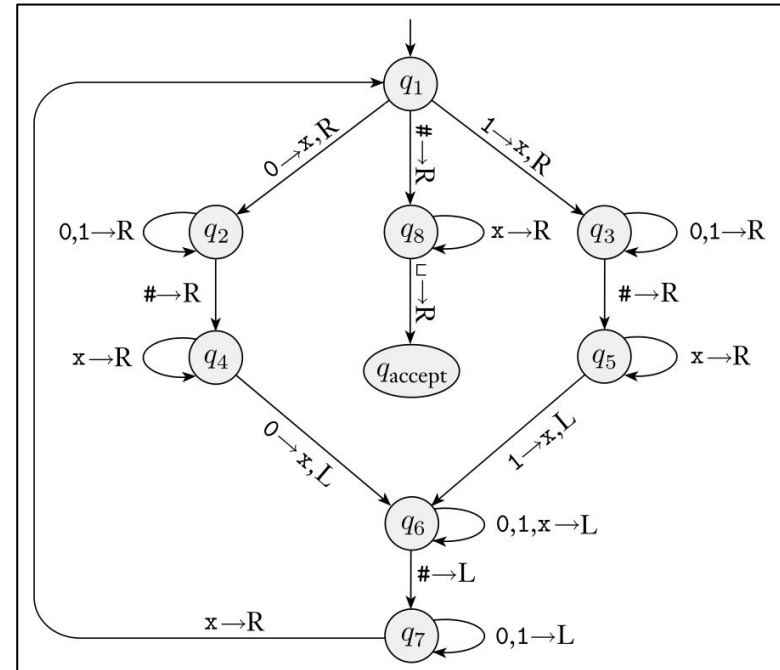
- Btw: We are constructing only deterministic machine which is the norm
- There is nondeterministic Turing machine which has the same power as the normal TM.



Examples for TM :

$w\#w$

- Btw: We are constructing only deterministic machine which is the norm
- There is nondeterministic Turing machine which has the same power as the normal TM.



Examples for TM :

WW

- How to construct the turing machine for this :



Examples for TM :

ww

- How to construct the turing machine for this :

**Algorithm for :
How to find the first word ?
Or
Middle of ww**

Examples for TM :

$$1^n x 1^m = 1^{nm}$$

- The language alphabet is : $\Sigma = \{1, x, =\}$
- Is this :
 - Regular language ?
 - Context free language ?

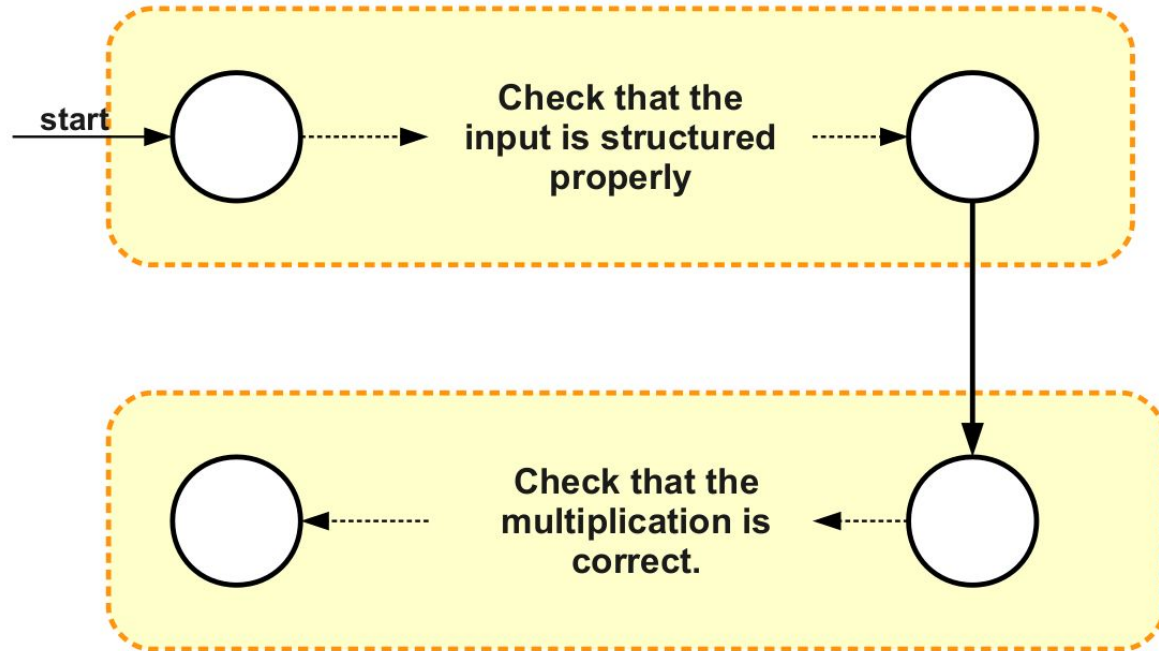
Examples for TM :

$$1^n \times 1^m = 1^{nm}$$

- Remember : Turing Machine is an Abstract computer with the same power as a real computer.
 - Subroutines can be created as a set of states to perform some business logic. The set of states have a single entry state and single exit state
 - Complex tasks can be performed by breaking tasks into subroutines

Examples for TM :

$$1^n \times 1^m = 1^{nm}$$



Examples for TM :

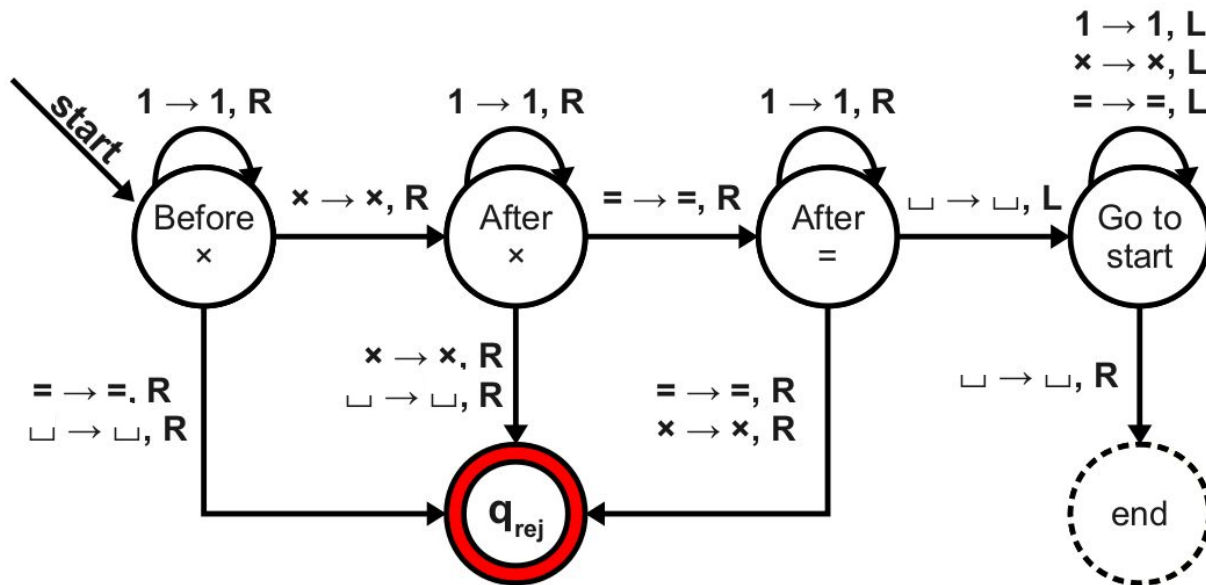
$$1^n x 1^m = 1^{nm}$$

- Let's build a “**subroutine**” TM for the language $1^* x 1^* = 1^*$

Examples for TM :

$$1^n x 1^m = 1^{nm}$$

- Let's build a “**subroutine**” TM for the language $1^* x 1^* = 1^*$



Examples for TM :

$$1^n \times 1^m = 1^{nm}$$

- How to make sure that : the number of 1s on the right hand side is correct ?
- Algorithm : ?

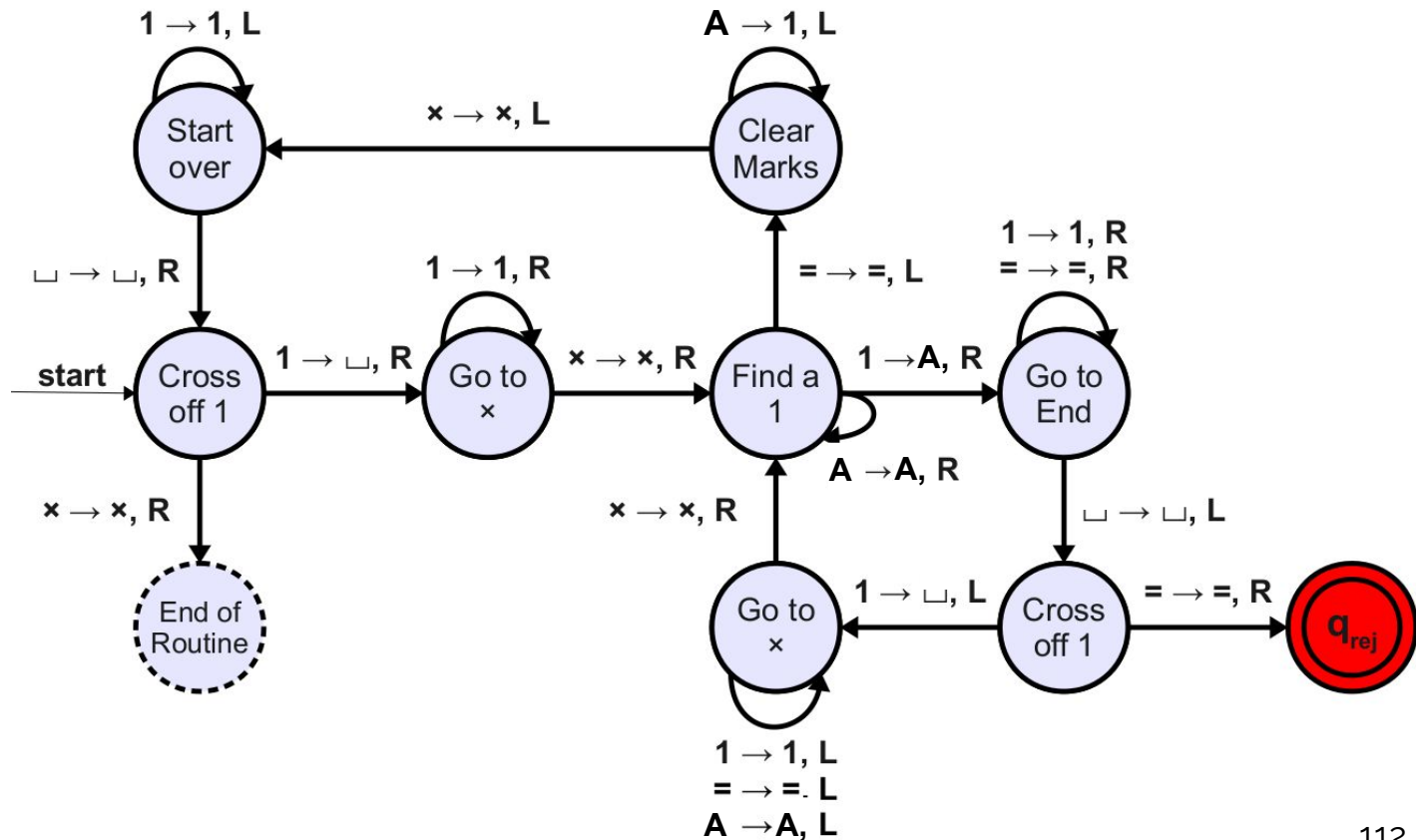
Examples for TM :

$$1^n \times 1^m = 1^{nm}$$

- Algorithm :
 - For the initial 1 found, replace with space.
 - Skip right till x,
 - For each 1 read, until =
 - Replace the 1 with A , move to find 1 on extreme **right**, replace with space
 - Get back till you find = and later A, and find 1 on the right.
 - If no 1 found , Replace all As with 1s.
 - Go to initial 1 on the extreme left. Repeat.
 - If no 1 is found before **x**, go to after =, there must be no 1 too.

Examples for TM :

$$1^n \times 1^m = 1^{nm}$$



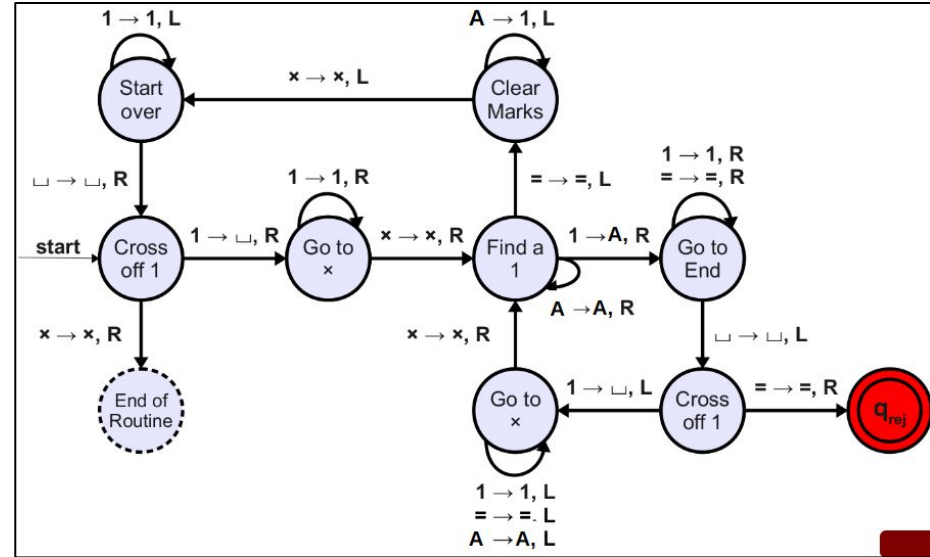
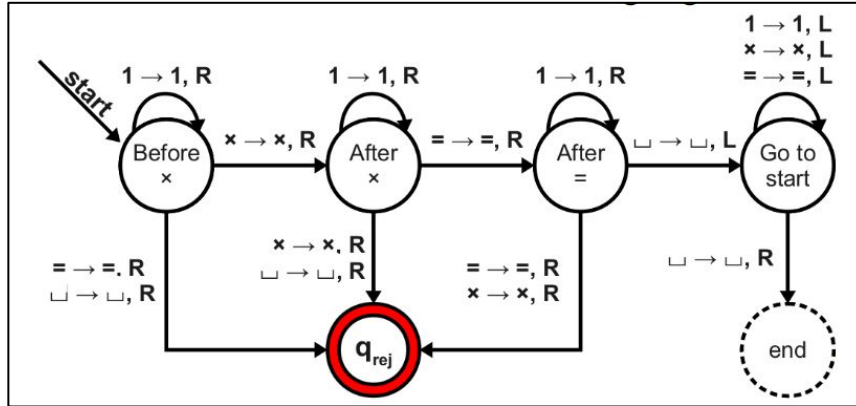
Examples for TM :

$$1^n \times 1^m = 1^{nm}$$

How to combine them

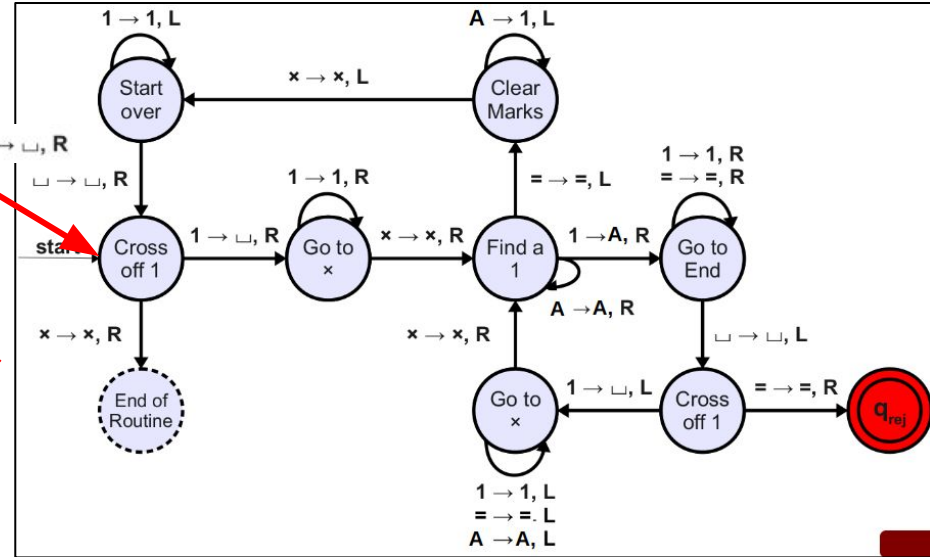
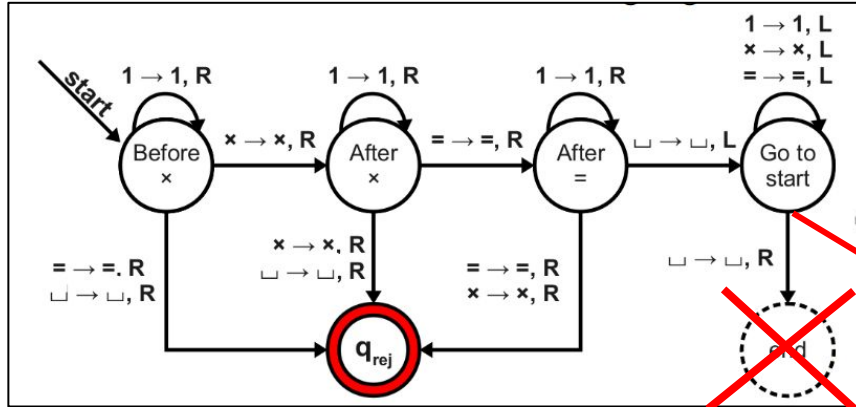
Examples for TM :

$$1^n \times 1^m = 1^{nm}$$



Examples for TM :

$$1^n \times 1^m = 1^{nm}$$



Formalism of Turing Machine

A *Turing machine* is a 7-tuple, $(Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$, where Q, Σ, Γ are all finite sets and

1. Q is the set of states,
2. Σ is the input alphabet not containing the *blank symbol* $\sqcup$,
3. Γ is the tape alphabet, where $\sqcup \in \Gamma$ and $\Sigma \subseteq \Gamma$,
4. $\delta: Q \times \Gamma \longrightarrow Q \times \Gamma \times \{L, R\}$ is the transition function,
5. $q_0 \in Q$ is the start state,
6. $q_{\text{accept}} \in Q$ is the accept state, and
7. $q_{\text{reject}} \in Q$ is the reject state, where $q_{\text{reject}} \neq q_{\text{accept}}$.

Classes of Languages for Turing Machine

- The collection of strings that M accepts is the language of M , or the language recognized by M
 - A language is called **Turing-recognizable** if some Turing machine recognizes it
 - Mainly : Accepting words that belong to the language.
 - For words not in the language:
 - Reject or Loop

Classes of Languages for Turing Machine



- **Turing-decidable** language or simply decidable if some Turing machine decides it
 - Halts and Accepts for words in the language
 - Halts and Reject for words not in the language
- Every Decidable language is also recognizable.

Classes of Languages for Turing Machine

- The collection of strings that M accepts is the language of M, or the language recognized by M
 - A language is called **Turing-recognizable** if some Turing machine recognizes it
 - Mainly : Accepting words that belong to the language.
 - For words not in the language:
 - Reject or **Loop**

Classes of Languages for Turing Machine



- **Terminologies:**

- Turing Recognizable is called a **recursively enumerable language** in some other textbooks.
- For turing decidable is called a **recursive language**

Classes of Languages for Turing Machine



- What about the following language :
 - $\{w\#z \mid \text{such that } w \text{ is not equal to } z \text{ and } w, z \in \Sigma^* \}$

Classes of Languages for Turing Machine

- What about the following language :
 - $\{w\#z \mid \text{such that } w \text{ is not equal to } z \text{ and } w, z \in \{0,1\}^* \}$
 - $\{x_1\#x_2\#x_3\#x_4\#x_5\# \dots \mid \text{such that } x_1, x_2, x_3, x_4, x_5 \dots \text{are all distinct} \}$

Revision :

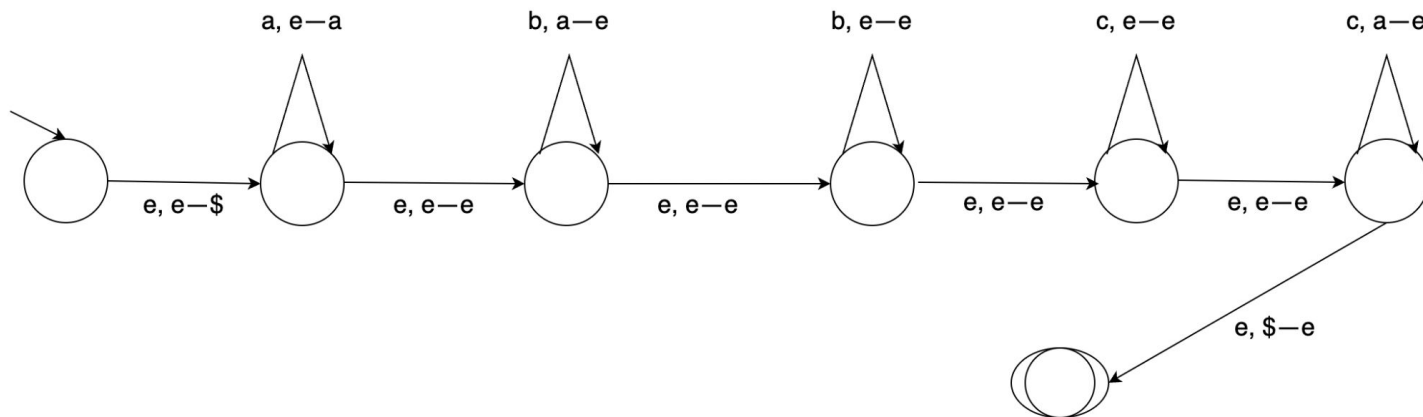
- DFA :
 - Construct the DFA (strictly not NFA) for the following language :
 - $(01+)^*$

Revision :

- PDA :
 - Let the language $L = \{ 0^n 1^m \}$ such that m is a multiple of n (or n is a factor of m)
 - Is the Language L a Context free language.
 - Examples : 0011, 001111, 00111111, 000111111111 ,,,

TD7 - Solutions

Draw a pushdown automaton for the following language $L = \{a^i b^j c^k \mid j+k \geq i\}$. Only one symbol is allowed to be pushed/popped to/from the stack at a time.

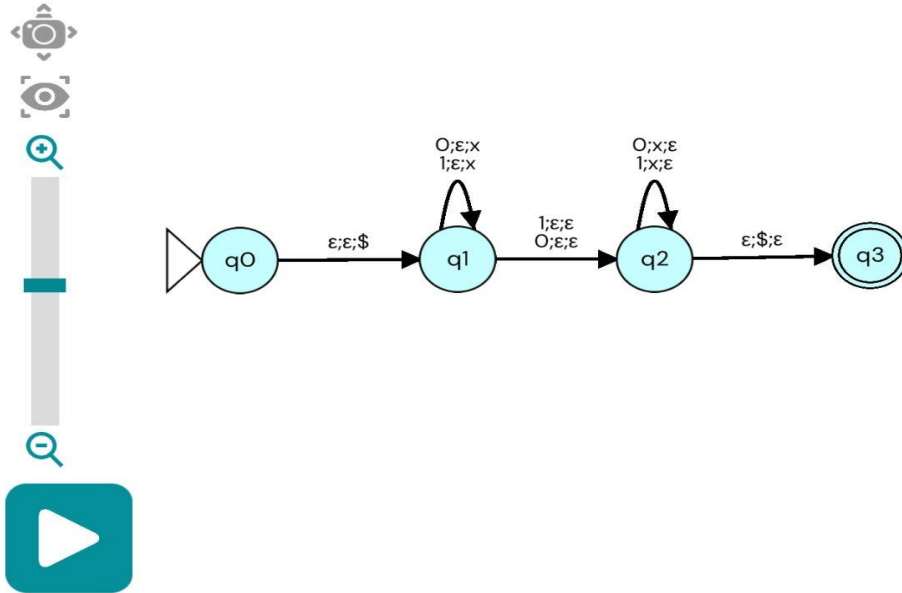


TD7 - Solutions

Give pushdown automata for the following languages:

$\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a } 0\}$

Simulation: Pending



The FSAM interface displays the following:

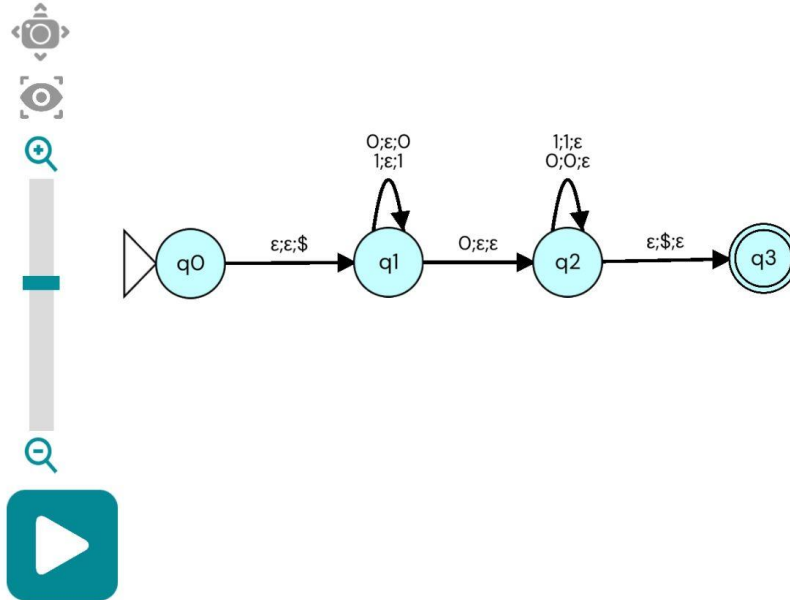
- Home icon (house)
- Accepted string: 100
- Rejected string: 0000
- Buttons: + (add), ? (help), and a play button at the bottom.

TD7 - Solutions

Give pushdown automata for the following languages:

$L = \{w \mid w = w^R, \text{ that is, } w \text{ is a palindrome and } |w| \text{ is odd}\}$

Simulation: Pending



The FSAM interface displays the following:

- A home icon and a menu icon (three horizontal lines) at the top left.
- A green box with the string "11011" and the status "Accepted".
- A red box with the string "0000" and the status "Rejected", with a close button (X) on the right.
- A plus button (+) below the rejected string.
- A pencil icon in a circle at the bottom right.
- The text "FSAM" written vertically on the left side of the interface.

TD7 - Solutions

Give pushdown automata for the following languages:

$$L = \{0^m 1^n : m \geq n\}$$

TD7 - Solutions

Give pushdown automata for the following languages:

$$L = \{u0w1 : u \text{ and } w \in \{0, 1\}^* \text{ and } |u| = |w|\}$$

TD7 - Solutions

Convert the following CFGs to an equivalent PDA:

a)

$$\begin{aligned} E &\rightarrow E+T \mid T \\ T &\rightarrow T \times F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

b)

$$\begin{aligned} S &\rightarrow aSa \mid bSb \mid aPb \mid bPa \\ P &\rightarrow aP \mid bP \mid \varepsilon \end{aligned}$$